Local order parameters for ice polymorphs: a unified comparison and practical guide

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Abstract

Molecular simulations of water and ice rely on local order parameters to tell solid from liquid, one ice polymorph from another, and to follow nucleation, growth and interfaces one molecule at a time. Dozens of such parameters have been proposed, from tetrahedral order and the local structure index to Steinhardt bond-orientational invariants, CHILL+, template matching and machine-learned descriptors, but they are usually compared with values taken from different codes, cutoffs and water models. We review these parameters in a single taxonomy organized by the structural information they use, and present a reproducible benchmark of selected parameters for crystalline ice and supercooled water that asks when they give reliable, transferable labels: how they survive thermal motion, whether their thresholds transfer between independent preparations and water models, how they depend on neighbor conventions, and how much crystal a labeler needs around a molecule. We compute 22 parameter families (39 scalar outputs) and three labelers with one open implementation and settings fixed in advance. The main benchmark is Ih, Ic, stacking-disordered ice and supercooled liquid in TIP4P/Ice at 220 K, with five independent preparations, extended to TIP4P/2005 and mW, to six high-pressure ices with pressure-matched liquids, and to constructed seeds; 18 ideal crystal structures serve as a structural reference. Locally averaged parameters, above all the averaged pair-entropy fingerprint and averaged q_6, separate ice from ambient supercooled liquid almost perfectly, and averaged q_3 and q_4 separate hexagonal from cubic ice. In the conditions tested, their thresholds transfer between independent preparations within 0.003, between protocols and the two TIP4P models within 0.03 and to mW within 0.01, and their scores change by at most 0.025 when the first-shell cutoff is moved between 3.2 and 3.8 Å. Distance- and density-based parameters transfer poorly between models and cutoffs, quenched configurations turn small density differences into spurious separations, and averaging beyond the first shell destroys the Ih/Ic signal of $\bar{q}_3$. Every pair of thermal high-pressure crystals is separated by some single parameter, but no single parameter separates every high-pressure ice from liquid at its own pressure, and none distinguishes proton-ordered from proton-disordered forms of one oxygen lattice. In constructed seeds, the fraction of molecules assigned the correct polytype rises with depth below the seed surface; CHILL+ reaches 90% at 5–7 Å and PTM about 2 Å deeper, while CHILL+ assigns many shallower molecules to its interfacial class. We conclude with recommendations tied to the tested conditions and a reporting checklist.

Contents

1 Introduction **3**

2 The ice landscape in brief **4**
2.1 Phase diagram and stability 4
2.2 Structural signatures relevant to local order 5
2.3 Amorphous ices, clathrates and liquid water 6

3 A taxonomy of local order parameters **6**
3.1 Common conventions 6
3.2 Radial and density-based parameters 7
3.3 Angular and tetrahedral parameters 7
3.4 Hydrogen-bond and topology-based parameters 8
3.5 Bond-orientational order 8
3.6 Template and symmetry-based parameters 8
3.7 Other recent parameters 9
3.8 Summary table 9

4 Descriptors and machine-learned approaches **10**
4.1 High-dimensional descriptors 10
4.2 Supervised classifiers 10
4.3 Unsupervised approaches 10
4.4 Data-driven collective variables 11

5 A unified benchmark **11**
5.1 Datasets 11
5.2 Separability metric 13
5.3 Ideal crystals 13
5.4 Thermal motion: averaging is what survives 14
5.5 How far do thresholds transfer? 16
5.6 High-pressure phases 19
5.7 Polytype recall against depth in constructed seeds 21
5.8 Application: nuclei along nucleation paths 23
5.9 Sensitivity to the neighbor convention 23
5.10 Computational cost 25
5.11 What the benchmark does not cover 26

6 Practical guide **26**
6.1 Software 27
6.2 Reproducibility checklist 28

7 Outlook **28**

1 Introduction

Water freezes into more crystalline structures than almost any other simple substance. Ice Ih, the form of snow and glaciers, is joined by more than twenty crystalline phases that are stable or metastable at higher pressures, lower temperatures or both, by several amorphous ices, and by empty and filled clathrate frameworks [1, 2, 3]. New phases are still being found: ice XIX was characterized in 2021 [4, 5], a metastable ice XXI was obtained from supercompressed water at room temperature [6], and ice XXII has been reported above 300 GPa [7]. Molecular simulation is the main tool for asking how these phases form: how ice nucleates from supercooled water [8, 9, 10, 11, 12], which polymorph or stacking sequence grows [13, 14, 11], and how ice behaves at interfaces and in confinement [15, 16].

Each of these questions requires deciding, for every molecule in every frame, what kind of local environment it is in. That decision is made with a *local order parameter* (LOP): a function of the positions of a molecule and its neighbors that takes characteristic values in each phase. A LOP can do three different jobs. As a *descriptor* its distribution fingerprints a phase or a state point. As a *classifier*, with a threshold or a set of rules, it labels individual molecules, from which nuclei, interfaces and phase fractions are built. As a *collective variable* (CV) it drives enhanced sampling, which additionally requires it to be continuous and differentiable in the coordinates [17, 18]. Success at one job does not imply success at another. A parameter whose distributions differ on average may still misclassify many molecules, and a good classifier may be useless as a CV.

The choice matters. Computed nucleation barriers and rates from seeding depend on the order parameter used to measure the nucleus [19, 20, 21], and the size of a nucleus or the width of an interface changes with the labeling criterion [22, 15]. Popular water order parameters are only weakly correlated with one another and probe different distortions of the hydrogen-bond network [23, 24]. Yet most comparisons take values from different codes, with different neighbor definitions, cutoffs, water models and reference structures, so that the parameter cannot be separated from the way it was computed. Machine-learned classifiers report accuracies above 99% [25, 26, 16], but on their own datasets, which makes them hard to compare with each other or with simpler parameters. Common-setting comparisons do exist: Doi and co-workers screened 493 order parameters and their combinations for Ih, Ic and liquid at coexistence and for the high-pressure triple points III–V–liquid and V–VI–liquid [27, 28, 29], and Shi and co-workers compared labelers on the quasi-liquid layer of ice [15]. What is less well established is whether such rankings are reproducible and transferable: how much of a separation survives thermal motion rather than only energy minimization, whether a threshold fitted on one trajectory works on an independently prepared one or on another water model, how strongly results depend on cutoffs and averaging radii, and how close to an interface a labeler can still assign the right polytype. These are the questions our benchmark addresses.

This review has four parts. Section 2 summarizes the ice landscape and the structural features a local parameter can and cannot see. Section 3 defines the physically motivated parameters in one notation, grouped by the information they use, and section 4 reviews descriptors, learned classifiers and data-driven collective variables. Section 5 reports a reproducible benchmark of selected local parameters for crystalline ice and supercooled water, in which every parameter

is computed by one open implementation, SolOrder, with settings fixed before any comparison. It examines thermal robustness, threshold transferability, neighbor conventions and the spatial limits of local classification; table 3 lists exactly what it covers. Section 6 turns the results into recommendations and a reporting checklist, and section 7 lists open problems.

Scope. We consider per-molecule parameters for bulk crystalline ices, empty clathrate frameworks and liquid water, computed from oxygen positions or full molecular geometries. We do not cover global crystallographic analysis, spectroscopic observables, electronic structure or superionic phases. Amorphous ices, machine-learned classifiers and collective variables are reviewed from the literature but not benchmarked.

2 The ice landscape in brief

2.1 Phase diagram and stability

Figure 1 shows the experimental phase diagram. Roman numerals now run to XXII; counting Ih and Ic separately gives more than twenty crystalline phases [1, 2, 30, 6, 7]. Two caveats apply to any count. The structure of ice XIX is disputed: it was assigned to a partially antiferroelectric ordering of ice VI in a $\sqrt{2} \times \sqrt{2} \times 1$ supercell [4, 5], while an alternative analysis finds HCl-doped ice VI to remain disordered and distorted [31]. And the numeral XXII has been used for two different structures, a high-pressure orthorhombic phase [7] and, in a preprint, a metastable phase grown from emulsified water at 250 K [32]. Beyond the numbered phases, ice VII becomes plastic (freely rotating molecules on a bcc lattice) at high temperature, as predicted by simulation [33] and observed by quasi-elastic neutron scattering [34], and superionic phases form at planetary conditions [35, 36, 37, 38].

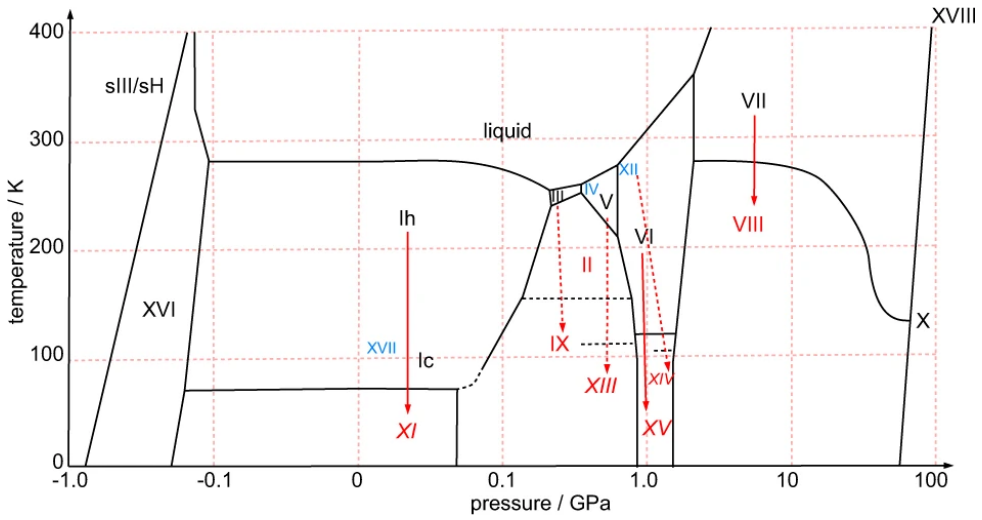


Figure 1: Experimental phase diagram of water. Black numerals: hydrogen-disordered phases; red: hydrogen-ordered; blue: metastable. Red arrows mark order–disorder transitions. Reproduced from Ref. [2] (CC BY 4.0).

Most phases come in pairs. Ih/XI, III/IX, V/XIII, VI/XV, VII/VIII and XII/XIV share an oxygen framework and differ in whether the protons are ordered; the ordered form is stable at

lower temperature [39, 40, 41]. Stacking adds a second kind of variety: ice I can stack its puckered layers hexagonally (Ih), cubically (Ic) or in any mixture, and ice that freezes from supercooled water is stacking-disordered (Isd) rather than cubic [14, 42, 43]. Stacking-disorder-free Ic has been made only indirectly, by emptying ice XVII or a hydrogen hydrate [44, 45]. Simulated critical nuclei are themselves stacking-disordered [11, 10, 46].

Simulations see the phase diagram of the water model, not that of water. Rigid nonpolarizable models reproduce its topology to different degrees [47, 48, 49, 50]. TIP4P/Ice melts at 272.2 K [51] and its full phase diagram has been mapped with enhanced sampling [52]; TIP4P/2005 melts about 20 K lower [53, 54] and its high-pressure diagram includes a plastic phase [33, 55]; the oxygen-only mW model melts at 274.6 K [56]. Machine-learned potentials trained on first-principles data now reproduce the phase diagram semi-quantitatively, including nuclear quantum effects [57, 58, 59, 60, 61]. A state point that is stable for one model can therefore be metastable, or a different phase, for another, and every simulation study of ice structure should state where its state point lies on the model's own phase diagram.

2.2 Structural signatures relevant to local order

Table 1 lists, for the phases used in our benchmark, the quantities a local parameter could exploit, computed from ideal GenIce structures [62, 63]. Three groups stand out. The low-density, fully tetrahedral frameworks (Ih, Ic, XVII and the clathrates) have exactly four oxygens within 3.5 Å, near-perfect tetrahedral order Q and a large gap to the fifth neighbor ($d_5 \approx 4.3$–4.4 Å). The intermediate-pressure phases (II, III, V, XII) keep four hydrogen bonds but fold the network so that non-bonded oxygens enter the first shell: four to six neighbors within 3.5 Å, Q of 0.79–0.86 and d_5 of 3.2–3.5 Å. Ice VI and the VII/VIII family consist of two interpenetrating networks, giving 11–14 neighbors within 3.5 Å and Q as low as 0.33–0.69, even though every molecule still donates and accepts two hydrogen bonds.

Table 1: Structural signatures of the crystalline phases in the benchmark. Density, coordination number (CN, O within 3.5 Å), mean tetrahedral order Q and fifth-neighbor distance d_5 are computed with SolOrder on ideal GenIce structures (ice Ic and Ih are shown with the same lattice spacing). Proton order: D disordered, O ordered; partner: the phase sharing the oxygen framework.

Phase	Space group	H	Partner	ρ (g cm^{-3})	CN	Q	d_5 (Å)
Ih	$P6_3/mmc$	D	XI	0.92	4.0	0.998	4.43
Ic	$Fd\bar{3}m$	D	–	0.92	4.0	0.998	4.43
II	$R\bar{3}$	O	–	1.18	5.5	0.83	3.23
III	$P4_12_12$	D	IX	1.17	4.5	0.86	3.53
V	$C2/c$	D	XIII	1.24	5.8	0.80	3.32
VI	$P4_2/nmc$	D	XV	1.37	11.5	0.69	3.36
VII	$Pn\bar{3}m$	D	VIII	1.60	14.0	0.40	2.90
VIII	$I4_1/amd$	O	VII	1.63	13.0	0.33	2.88
XII	$I\bar{4}2d$	D	XIV	1.44	9.6	0.79	3.28
XVII	$P6_122$	D	–	0.88	4.0	0.97	4.29
sI (empty)	$Pm\bar{3}n$	D	–	0.80	4.0	0.98	4.40

What a local parameter cannot see follows from the same table. Proton-order partners share an oxygen sublattice, so any oxygen-only parameter can separate them only through the small lattice distortions that accompany ordering. Ih and Ic have identical first and nearly identical

second shells; they differ in the relative orientation of neighboring tetrahedra (staggered versus eclipsed bonds) and in the third shell, so a parameter must either encode bond orientations beyond the first shell or average over neighbors. Rings are equally uninformative here: both Ih and Ic consist of six-membered rings only.

2.3 Amorphous ices, clathrates and liquid water

Low-density (LDA), high-density (HDA) and very-high-density amorphous ices are glassy solids whose local structure resembles that of liquids at similar density [64, 65, 66, 67], and a medium-density amorphous ice obtained by ball-milling Ih falls in the gap between LDA and HDA [68]. Clathrate hydrates build polyhedral cages from five- and six-membered rings around guest molecules; their empty frameworks (ices XVI and XVII among them) are tetrahedral like Ih [69, 70]. Supercooled liquid water, finally, becomes increasingly tetrahedral on cooling and has been described as a fluctuating mixture of low- and high-density local structures [71, 72, 73, 74]; it contains transient ice-like environments [75, 76]. These three are the hardest cases for local parameters, because they share the tetrahedral first shell of crystalline ice and differ from it mainly in medium-range order.

3 A taxonomy of local order parameters

We group physically motivated parameters by the information they use: distances (radial), angles within the first shell, the hydrogen-bond network, bond orientations expressed in spherical harmonics, and comparison with a template. Table 2 summarizes them with the settings used in the benchmark.

3.1 Common conventions

Throughout, $\mathbf{r}_{ij} = \mathbf{r}_j - \mathbf{r}_i$ is the minimum-image vector between oxygens i and j and r_{ij} its length. Every parameter needs a neighbor set $N_b(i)$, and the choice is part of the definition: a radial cutoff (3.5 Å for the first shell of ambient ices, near the first minimum of g_{OO}), a fixed number of nearest neighbors (four for tetrahedral parameters), Voronoi neighbors, or hydrogen-bond partners. These choices give different results [77], especially at high pressure, where non-bonded oxygens enter the first coordination shell. Many parameters exist in a *locally averaged* form,

$$\bar{X}(i) = \frac{1}{|N_b(i)| + 1} \sum_{j \in \{i\} \cup N_b(i)} X(j), \tag{1}$$

introduced for bond-orientational order by Lechner and Dellago [78] (who average the complex q_{lm} vectors rather than the scalar, section 3.5). Averaging trades spatial resolution for noise suppression, and as we will see it is the single most important ingredient at finite temperature. Finally, a parameter used as a CV must be a continuous, differentiable function of the coordinates. Hard cutoffs, nearest- neighbor ranks and integer counts are not; switching functions and Gaussian broadening help, but only if neighbor membership and averaging are smooth as well.

3.2 Radial and density-based parameters

The *coordination number* counts oxygens within r_c; its smooth form replaces the step function with a switching function,

$$\mathrm{CN}_s(i) = \sum_{j \neq i} \frac{1 - (r_{ij}/r_0)^n}{1 - (r_{ij}/r_0)^m}, \qquad n = 12,\ m = 28. \tag{2}$$

The *fifth-neighbor distance* d_5, the distance to the fifth-nearest oxygen, measures how far the second shell is from the first and is large in low-density, tetrahedral environments [79, 80]. The *local structure index* [81, 82] sorts the distances $r_1 < r_2 < \cdots < r_n < 3.7\ \mathring{A} < r_{n+1}$ and measures the variance of the gaps $\Delta_k = r_{k+1} - r_k$,

$$\mathrm{LSI}(i) = \frac{1}{n} \sum_{k=1}^{n} \left(\Delta_k - \bar{\Delta}\right)^2, \tag{3}$$

which is large when a clear gap separates the first and second shells. The Voronoi cell of an oxygen gives a local density $1/V$ and a shape measure, the asphericity $\eta = A^3/(36\pi V^2)$, which is 1 for a sphere and 2.25 for ice Ih [83, 84, 85]. The *pair-entropy fingerprint* [86, 87] projects the two-body excess entropy onto each particle,

$$s(i) = -2\pi\rho k_B \int_0^{r_m} \left[g_m^i(r) \ln g_m^i(r) - g_m^i(r) + 1\right] r^2\, dr, \tag{4}$$

where $g_m^i(r)$ is a Gaussian-broadened radial distribution centered on i and r_m is a cutoff of about 5 Å. Its local average $\bar{s}$ was designed as a crystallization CV and uses the whole radial structure out to the second shell.

3.3 Angular and tetrahedral parameters

The *orientational tetrahedral order* [88, 89],

$$Q(i) = 1 - \frac{3}{8} \sum_{j=1}^{3} \sum_{k=j+1}^{4} \left(\cos\psi_{jik} + \frac{1}{3}\right)^2, \tag{5}$$

uses the angles ψ_{jik} between the four nearest oxygens; it is 1 for a regular tetrahedron and 0 on average for an ideal gas. The *translational tetrahedral order* $S_k = 1 - \frac{1}{3}\sum_{k=1}^{4}(r_k - \bar{r})^2/(4\bar{r}^2)$ measures the variance of the four nearest distances [88]. A family of parameters developed for structure classification by Takahashi, Doi and co-workers [90, 27, 91] uses all angles θ_{jik} in the first shell. The bond-angle parameter averages $\cos^{n_1}(n_2\theta_{jik} + \phi)$ [92, 91]; the angular Fourier series parameter averages $f(\min(r_{ij}, r_{ik}))\cos(a\theta_{jik})$ [93, 91]; the angle-histogram parameters H_1–H_3 are Fourier amplitudes of the histogram of θ_{jik} [91]; and the common-neighborhood parameters A_1–A_3 sum vectors to neighbors shared by i and each of its neighbors [94, 95, 91].

3.4 Hydrogen-bond and topology-based parameters

These parameters need hydrogen positions or a hydrogen-bond network. We use the geometric criterion of Luzar and Chandler: an O–O distance below 3.5 Å and an H–O$\cdots$O angle below 30° [96]. N_{HB} is the number of donated and accepted bonds. The four-body parameter $F_4 = \langle \cos 3\phi \rangle$ averages over neighbors the H–O$\cdots$O–H torsion formed with each molecule's outer hydrogens [97]; it is about -0.4 in ice, -0.04 in the liquid and 0.7 in clathrate hydrates [97, 98]. The selection of the outer hydrogen matters: choosing it per molecule from the opposite oxygen is the correct form, and we found and fixed an implementation that did otherwise. The Russo–Tanaka ζ is the distance to the nearest non-hydrogen-bonded oxygen minus that to the farthest hydrogen-bonded one [71]; it is positive in locally favored, tetrahedral structures. Ring statistics count the smallest rings of the hydrogen-bond network passing through each molecule [99, 100, 101]. Different ring definitions give different results [102], and in small periodic cells paths that wrap around the box must be rejected, or they are counted as rings.

3.5 Bond-orientational order

Steinhardt parameters [103] expand the directions to the neighbors in spherical harmonics,

$$q_{lm}(i) = \frac{1}{|N_b(i)|} \sum_{j \in N_b(i)} Y_{lm}(\hat{\mathbf{r}}_{ij}), \qquad q_l(i) = \sqrt{\frac{4\pi}{2l+1} \sum_{m=-l}^{l} |q_{lm}(i)|^2}, \tag{6}$$

and build the third-order invariant $w_l = \sum_{m_1+m_2+m_3=0} \begin{pmatrix} l & l & l \\ m_1 & m_2 & m_3 \end{pmatrix} q_{lm_1} q_{lm_2} q_{lm_3}$ and its normalized form $\hat{w}_l = w_l / (\sum_m |q_{lm}|^2)^{3/2}$. Lechner and Dellago average the vectors, $\bar{q}_{lm}(i) = \sum_{j \in \{i\} \cup N_b(i)} q_{lm}(j) / (|N_b(i)| + 1)$, and form $\bar{q}_l$ from them [78]; this picks up the orientation of the second shell. Odd l matter for water: $l = 3$ captures tetrahedral symmetry, and $\bar{q}_3$ and $\bar{q}_4$ distinguish Ih from Ic [104, 27]. The ten Wolde–Frenkel *crystalline connections* count neighbors j with normalized dot product $\mathbf{q}_6(i) \cdot \mathbf{q}_6(j) > 0.5$ [105]. For water, Reinhardt and co-workers designed Steinhardt-based parameters able to drive ice growth [106], Haji-Akbari and Debenedetti labeled molecules with $q_6 > 0.5$ within 3.2 Å in their direct nucleation-rate calculation [10], and seeding studies tune a $\bar{q}_6$ threshold by a mislabeling criterion [19].

The CHILL algorithm [107] uses the correlation $c(i,j) = \mathbf{q}_3(i) \cdot \mathbf{q}_3(j) / (|\mathbf{q}_3(i)||\mathbf{q}_3(j)|)$ for each bond to the four nearest neighbors. Bonds with $c < -0.8$ are staggered and bonds with $-0.2 < c < -0.05$ eclipsed; four staggered bonds mean cubic ice and three staggered plus one eclipsed hexagonal ice, with further rules for interfacial ice. CHILL+ [108] widens the eclipsed window to $-0.35 < c < 0.25$, uses neighbors within 3.5 Å, and adds clathrate classes for molecules with four eclipsed bonds. It was designed to detect hexagonal ice up to its melting point.

3.6 Template and symmetry-based parameters

The centrosymmetry parameter sums $|\mathbf{r}_{ij} + \mathbf{r}_{ik}|^2$ over opposite pairs of the 12 nearest neighbors and vanishes for a centrosymmetric environment [109]. The neighbor-distance parameter D averages products of functions of the three distances in each neighbor triangle [91]. Polyhedral

template matching (PTM) finds the template (here hexagonal or cubic diamond) with the lowest root-mean-square deviation after optimal rotation and scaling, and accepts it below an RMSD cutoff [110]; the related identify-diamond method uses common neighbor analysis of the second shell [111, 112]. The local-order metric of Martelli and co-workers measures the best overlap of a neighborhood with a reference lattice [113, 114].

3.7 Other recent parameters

Several recent parameters are not in our benchmark: an O–O–O–O dihedral parameter that separates Ih, Ic and high- and low-density liquid with Bayesian classification [115], the hydrogen-bond-network indicator Ψ [116], and an environment-similarity CV for oxygen-only ice growth [117]. Supervised screens over hundreds of candidate parameters and their combinations [90, 27, 28, 29, 118] and graph-network-designed parameters [119] have also produced task-specific sets.

3.8 Summary table

Table 2 lists every parameter in the benchmark with the settings used there. "Smooth form" marks a parameter defined through a switching function or Gaussian broadening. In our implementation neither marked parameter is a tested CV: both are truncated at the 5 Å search cutoff (where the switching function is small but not zero) and use hard neighbor membership.

Table 2: Local order parameters in the benchmark, with the settings used there. All O–O first-shell cutoffs are 3.5 Å. H: needs hydrogen positions.

Parameter	Group	Neighbors	H	Smooth form	Ref.
CN	radial	O within 3.5 Å			–
Smooth CN	radial	switching function, $r_0 = 3.5$ Å		✓	–
d_5	radial	fifth-nearest O			[80]
LSI	radial	O within 3.7 Å and next			[81]
Voronoi density, asphericity	radial	periodic Voronoi cell			[83, 84]
Entropy s, $\bar{s}$	radial	$g(r)$ to 5 Å; average to 5 Å		✓	[86, 87]
Q, $\bar{Q}$	angular	4 nearest O; average 3.5 Å			[89]
S_k	angular	4 nearest O			[88]
Bond angle B	angular	O within 3.5 Å			[92, 91]
Angular Fourier ($a = 3$)	angular	O within 3.5 Å			[93, 91]
Angle histogram H_1–H_3	angular	O within 3.5 Å			[91]
A_1–A_3	angular	O within 3.5 Å			[95, 91]
F_4	H-bond	O within 3.5 Å, outer H	✓		[97]
N_{HB}	H-bond	3.5 Å, 30°	✓		[96]
ζ	H-bond	H-bond partners; O to 5 Å	✓		[71]
Ring fractions (4–8), mean	H-bond	H-bond network, rings ≤ 10	✓		[99, 100]
q_3, q_4, q_6	bond-orient.	O within 3.5 Å			[103]
$\bar{q}_3$, $\bar{q}_4$, $\bar{q}_6$	bond-orient.	averaged over 3.5 Å			[78]
$\hat{w}_4$, $\hat{w}_6$	bond-orient.	O within 3.5 Å			[103]
Crystalline connections	bond-orient.	$\mathbf{q}_6$ dot product > 0.5			[105]
Centrosymmetry	template	12 nearest O			[109]
Neighbor distance D	template	O within 3.5 Å			[91]
CHILL / CHILL+	labeler	4 nearest / 3.5 Å, $\mathbf{q}_3$			[107, 108]
PTM	labeler	diamond templates, RMSD < 0.1			[110]

4 Descriptors and machine-learned approaches

4.1 High-dimensional descriptors

Machine-learning methods replace one scalar with a vector that describes the whole neighborhood. A useful descriptor is invariant to translation, rotation and permutation of identical atoms, changes smoothly with the coordinates, is complete enough to distinguish different environments, and is cheap to compute [120, 121]. Atom-centered symmetry functions sum radial and angular terms over neighbors [122]. The smooth overlap of atomic positions (SOAP) expands a Gaussian-smeared neighbor density in radial functions and spherical harmonics and forms the rotationally invariant power spectrum $p_{nn'l} = \sum_m c_{nlm} c_{n'lm}$ [93, 123]; Steinhardt q_l is closely related to it, retaining a single radial channel [120]. Many-body tensor representations encode distributions of distances and angles [124]. Libraries such as DScribe [125] make these descriptors easy to compute. For water, SOAP shows that the liquid contains local environments resembling a wide range of ice phases [75], that classical rigid models contain short-lived ice-like environments [76], and that SOAP and standard order parameters do not tell the same story about the liquid [126, 24].

4.2 Supervised classifiers

Supervised networks are trained on molecules of known phase. Geiger and Dellago trained a neural network on symmetry functions to identify liquid and several ice phases [127]. DeepIce feeds the Cartesian coordinates of the nearest neighbors to a deep network and was applied to ice surfaces, where it outperformed Steinhardt, tetrahedral and template labels [128]. A PointNet-based network reached 99.1% accuracy over liquid, five ice phases and sI and sII hydrates using oxygen positions alone, about 99.6% with hydrogens, with most errors between sI and sII; with interfacial molecules included its accuracy fell to about 92% [25]. GCIceNet, a graph convolutional network on the hydrogen-bond graph with order parameters as node features, reported 99.8% test accuracy over nine TIP4P/Ice classes [26], and an edge-conditioned graph network 99.9% over nine phases, including the interface and water in nanotubes [16]. Score-based denoising can first remove thermal noise and then classify with simple parameters [129], and a variational autoencoder on SOAP features identifies ice phases without labels in its latent space [130].

These accuracies are not directly comparable. Each refers to its own classes, water model, temperature and train/test split; most test molecules come from the same trajectories as the training molecules; and accuracy on bulk phases says little about interfaces, which is where the classification matters most. The benchmark in section 5 shows that, with averaging, simple scalars already separate ambient ice from liquid with more than 99% balanced accuracy on held-out runs, so a learned model has to be judged on harder tasks: interfaces, high-pressure ice against dense liquid, and transfer to other water models.

4.3 Unsupervised approaches

Unsupervised methods look for structure in descriptor space without labels. Autoencoders and Gaussian mixtures on bond-order parameters recover crystalline and hidden local structures in colloidal systems [131, 132]; persistent-homology descriptors with clustering follow nucleation in

metals [133, 134]; and a generalized convex hull with sketch-map projection organized nearly 75,000 hypothetical ice structures derived from zeolite networks and recovered the known phases among them [135]. Near the liquid–liquid critical point of a water model, local descriptors show no two-state structure while non-local ones reveal low- and high-density domains [136]. Unsupervised maps are descriptive. Turning a cluster into a phase label still requires reference structures, and the clusters depend on the descriptor and its cutoffs.

4.4 Data-driven collective variables

Enhanced sampling of nucleation needs a CV that is differentiable and that follows the slow structural change. Hand-built CVs for ice include Steinhardt-based parameters [137, 106], the averaged pair entropy [87], an environment-similarity kernel [117], and combinations tuned for crystallization [138, 46]. Data-driven CVs learn a low-dimensional combination of many descriptors, by discriminant analysis between basins (Deep-LDA) [139] or by the slowest modes of a biased trajectory (Deep-TICA) [140], and are typically combined with OPES [141]. The mlcolvar library exports such models to PLUMED [142]. Graph neural networks can learn nucleation CVs from Steinhardt features [143] or directly from coordinates [144], although neither has yet been tested on ice. In parallel, machine-learned potentials of first-principles quality [145, 146, 147, 148, 149] have brought nucleation studies to ab initio accuracy [150, 151, 12, 152]; they all still rely on a local order parameter to define the nucleus.

5 A unified benchmark

All parameters in table 2 were computed with one implementation (SolOrder, MIT license), one set of settings fixed before any comparison, and one analysis pipeline. Differences between parameters therefore reflect the parameters, not codes or cutoffs; the effect of the water model was measured separately. Before the benchmark, every parameter was checked against independent references: q_l against the Legendre addition formula (to 2×10^{-15}), direct geometric formulas for tetrahedral, radial and LSI parameters, MDAnalysis torsions for F_4, OVITO for CHILL+ (exact agreement on all GenIce phases tested), and serial against parallel execution on 366 frames from 44 nucleation trajectories.

5.1 Datasets

Ideal crystals. Eighteen structures were generated with GenIce2 [62, 63]: Ih, Ic, II, III, IV, V, VI, VII, VIII, IX, XI, XII, XIII, XIV and XVII, and the empty sI, sII and sH clathrate frameworks. Each proton-disordered phase was generated from five random seeds (89,100 molecular environments). Ice XVI was excluded because GenIce gives it the same oxygen positions as the empty sII framework. Every atom was displaced by Gaussian noise of 0.02 Å, about twice the coordinate precision of the files, so that two phases count as separable only at a stated resolution. Five pairs are proton order–disorder partners: Ih/XI, III/IX, V/XIII, VII/VIII and XII/XIV.

Seed test (run A). Bulk Ih, Ic and stacking-disordered ice, and spherical Ih, Ic and stacking-disordered seeds of 50, 100 and 200 molecules in liquid, were simulated with TIP4P/Ice [51] at 220 K at constant volume. Each of the twelve systems contributes ten thermal frames and the same frames after energy minimization (834,040 environments). Every molecule carries a label from the construction: seed membership, depth below the seed surface, and hexagonal or cubic stacking of its layer. The liquid reference consists of molecules more than 8 Å beyond the seed radius.

Replicas (run B) and independent preparations (runs C1–C5). Run B repeats bulk Ih, Ic, stacking-disordered ice and liquid (4096 molecules) at 220 K, and Ih and Ic at 250 K, from the same structures, with 60 frames 5 ps apart (1,643,520 environments). Runs C1–C5 start from new structures: Ih and Ic get new proton configurations, generated by ice-rule loop moves [153, 154] with the net dipole below 2% of full polarization; stacking-disordered ice gets a new random stacking sequence; each liquid is remelted at 300 K and re-equilibrated for 1 ns at 220 K (8,217,600 environments). Crystal oxygens stay within 0.35–0.41 Å of their minimized sites. The 220 K liquid is slow (about 3 Å per ns) and still aging slightly, but CHILL+ labels about 10% of its molecules as ice or interfacial ice with no trend.

Water models (run D). TIP4P/2005 [53] was run on the C1–C3 structures at 220 K (the same temperature) and at 200 K (about the same undercooling), together with a TIP4P/Ice control under the same protocol, so that moving a threshold from run C to the control measures the protocol effect and moving it from the control to TIP4P/2005 the model effect. Each system was equilibrated at 1 bar (anisotropic Monte Carlo barostat for crystals) and then sampled at constant volume (9,953,280 environments). The oxygen-only mW model [56], implemented in OpenMM [155] and checked against an independent energy calculation to 3×10^{-15}, was run on the C1–C3 oxygen positions at 220 K, close to the TIP4P/Ice undercooling (3,548,160 environments). Ih densities are 0.925 (TIP4P/2005), 0.912 (TIP4P/Ice) and 0.979 $\mathrm{g\,cm^{-3}}$ (mW).

High-pressure phases (run E). Ices II and III at 0.3 GPa, V at 0.5 GPa, VI at 1.0 GPa, and VII and VIII at 3.0 GPa were generated with GenIce2 (two seeds, every periodic height at least 24 Å), and liquid at each of these pressures (0.3, 0.5, 1.0 and 3.0 GPa) from the 4096-molecule liquid, remelted at 300 K and the target pressure, all with TIP4P/Ice at 220 K. Crystals were equilibrated for 200 ps at pressure (flexible barostat for the triclinic cells) and sampled as above (5,432,640 environments). All phases were retained: oxygens moved 0.25–0.45 Å over 300 ps, and the mean cosine between each molecule's dipole direction at the first and last frame stayed at 0.96, including the proton-ordered II and VIII. Densities were 1.18 (II), 1.15 (III), 1.26 (V), 1.35 (VI), 1.56 (VII) and 1.57 $\mathrm{g\,cm^{-3}}$ (VIII). Ice VII is metastable relative to VIII at 220 K and stays disordered only because rigid-water simulations do not order protons on this time scale.

Every thermal frame has a matching energy-minimized (quenched) frame, which removes thermal vibration while keeping the structure. Table 3 summarizes the coverage and the role of each dataset. The benchmark contains about 3.0×10^7 molecular environments, but that number measures computational coverage, not evidence: molecules in one frame, and frames of one trajectory, are strongly correlated. The inferential weight rests on the independent units,

which are five preparations for the ambient phases at 220 K, three per water model and two per high-pressure phase.

Table 3: Coverage of the benchmark. Prep.: independent preparations (new proton configurations, stacking sequences or remelted liquids). Frames: thermal frames per run, each with a quenched copy. Everything not listed, including amorphous ices, thermal clathrates, ices IV, IX and XI–XVII at finite temperature, three-site water models and other temperatures and pressures, is outside the benchmark.

Dataset	Phases	Model	T, P	Prep.	Frames	Role
Ideal	18 GenIce structures	none	–	5 seeds[a]	1	structural reference
Run A	bulk Ih, Ic, Isd; seeds of 50–200 molecules in liquid	TIP4P/Ice	220 K, 1 bar	1	10, 1 ps	interface depth, labelers
Run B	Ih, Ic, Isd, liquid	TIP4P/Ice	220 K (Ih, Ic also 250 K), 1 bar	1[b]	60, 5 ps	held-out transfer
C1–C5	Ih, Ic, Isd, liquid	TIP4P/Ice	220 K (Ih, Ic also 250 K), 1 bar	5	60, 5 ps	main benchmark
Run D	Ih, Ic, Isd, liquid	TIP4P/2005; TIP4P/Ice NPT control; mW	220 K (TIP4P/2005 also 200 K[c]), 1 bar	3	60, 5 ps	model and protocol transfer
Run E	II, III, V, VI, VII, VIII; liquid at each pressure	TIP4P/Ice	220 K, 0.3–3.0 GPa	2	60, 5 ps	high-pressure extension
Sensitivity	Ih, Ic, liquid (C1–C3)	TIP4P/Ice	220 K, 1 bar	3	12, 25 ps	neighbor conventions
Nucleation	44 trajectories	TIP4P/Ice, mW	various	–	366 total	qualitative application

[a] For proton-disordered phases; ordered phases repeat. [b] Same starting structures as run A. [c] Ih, Ic and liquid.

5.2 Separability metric

For a scalar parameter and two populations A and B we use the two-sample Kolmogorov–Smirnov statistic $D_{AB} = \sup_x |F_A(x) - F_B(x)|$, where F_A and F_B are the empirical cumulative distributions over the molecules of each population. D needs no binning, which matters for the near-delta distributions of ideal crystals, and $(1 + D)/2$ is exactly the balanced accuracy of the best single threshold chosen on the same data. We report it as an *oracle score*, between 0.5 (no separation) and 1 (perfect separation). Because the threshold is optimized on the evaluated molecules, it is not a predictive accuracy. We therefore also report *held-out* scores: the balanced accuracy of a threshold and direction fitted on one run and applied to another, with missing values counted as errors. Molecules within one frame are correlated, so bootstrap intervals resample frames (blocks of five frames, 25 ps, in run B) rather than molecules. Such intervals describe variation within one trajectory and are therefore narrow; the spread between independent preparations is the better measure of uncertainty and is the one we quote. Labelers (CHILL, CHILL+, PTM) are scored against the construction labels.

5.3 Ideal crystals

Of the 153 pairs of ideal crystals, 150 are separated at an oracle score of at least 0.95 by at least one parameter (fig. 2). The three exceptions are Ih/XI (best 0.87, smooth CN), V/XIII

(best 0.83, $\bar{q}_4$) and sI/sII (best 0.92). The first two are proton order–disorder partners whose oxygen frameworks barely differ, and no hydrogen-aware parameter (F_4, ζ, N_{HB}, ring fractions) separates them either. The other partner pairs are separable only because GenIce's ordered cells are slightly distorted: III/IX by $\bar{s}$ alone, VII/VIII by 17 parameters and XII/XIV by five. The most broadly useful parameters are $\bar{s}$ (93% of pairs at 0.95 or above), Voronoi asphericity (90%), Voronoi density and $\bar{Q}$ (89%). N_{HB} separates no pair, because every molecule in every ideal crystal has four hydrogen bonds; A_3 separates none either.

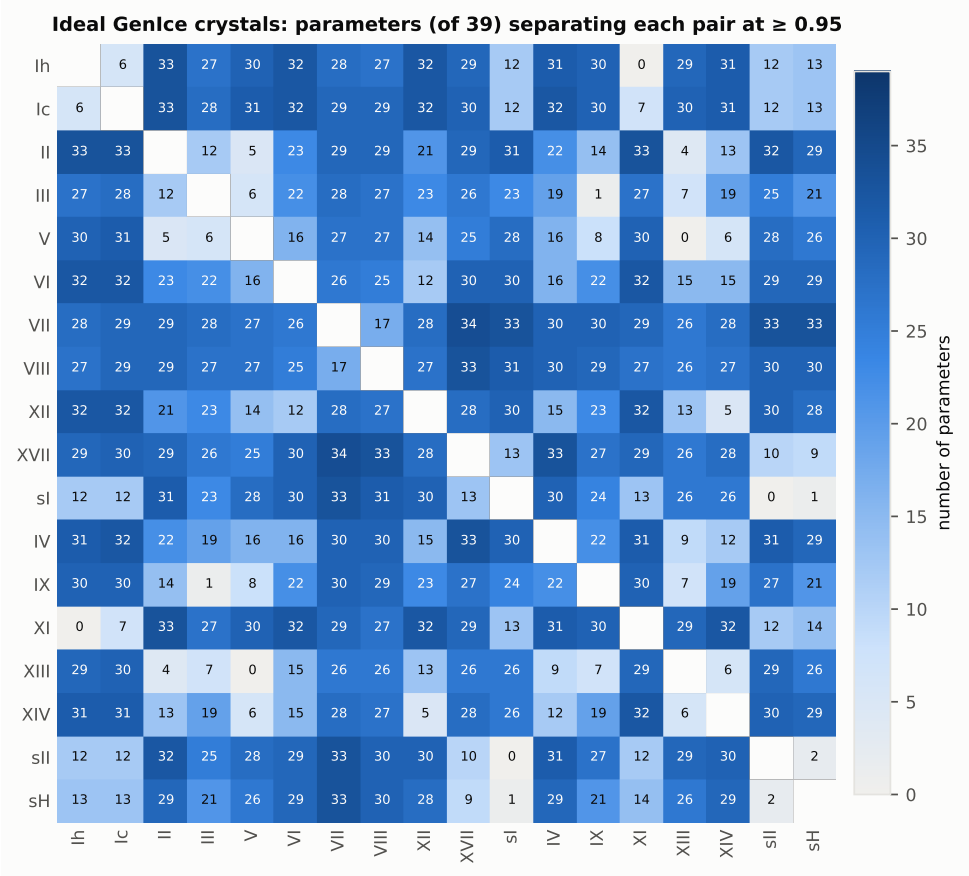


Figure 2: Number of parameters (of 39) that separate each pair of ideal crystals with an oracle score of at least 0.95.

5.4 Thermal motion: averaging is what survives

At 220 K the picture changes (fig. 3). For ice against liquid, the best parameters are locally averaged: $\bar{s}$ (1.00), $\bar{q}_6$ (0.99) and $\bar{Q}$ (0.98), compared with 0.85, 0.78 and 0.84 for their unaveraged forms. The six-ring fraction (0.99) is the best unaveraged parameter, since ring statistics already span several molecules. The cutoff coordination number (0.62) and N_{HB} (0.53) are nearly useless: supercooled liquid water is almost fully four-coordinated and hydrogen-bonded. Energy minimization restores most of the separation removed by thermal motion: d_5, LSI and ζ rise from 0.83–0.84 to 1.00.

Hexagonal and cubic ice are separated at 220 K only by $\bar{q}_3$ (1.00), $\bar{q}_4$ (1.00), F_4 (0.94), crystalline connections (0.93), $\bar{q}_6$ (0.78) and centrosymmetry (0.72); every other parameter is at 0.61 or below, including ring statistics, which cannot distinguish them. $\bar{s}$ separates Ih from Ic

perfectly in ideal crystals but only at 0.61 at 220 K. At 250 K the order is unchanged ($\bar{q}_3$ 1.00, $\bar{q}_4$ 0.99, F_4 0.91, crystalline connections 0.88). These results agree with earlier findings that $\bar{q}_4$ separates Ih from Ic at the ice–water interface [104] and that low-l averaged parameters are the best pair in a supervised screen [27].

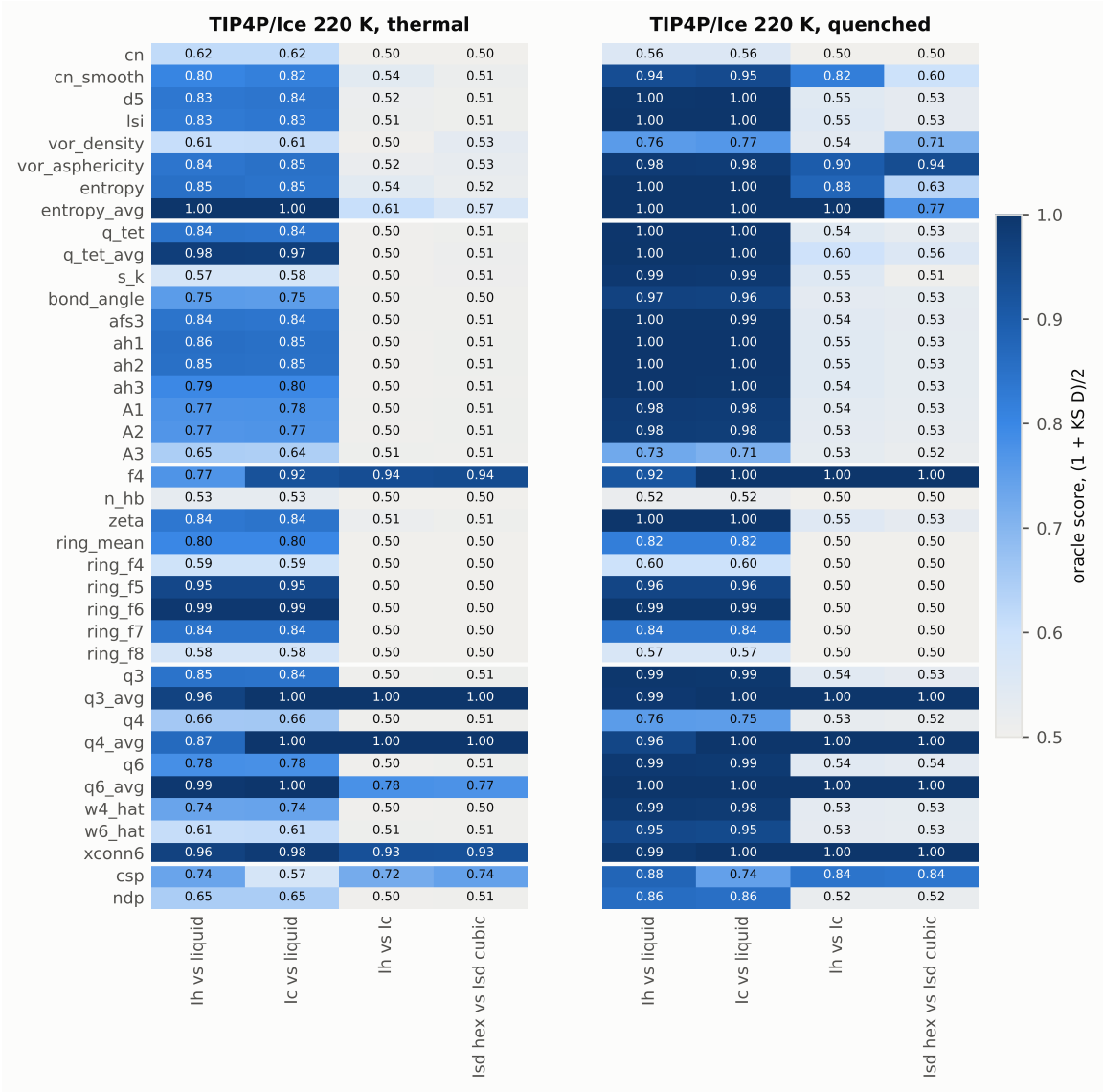


Figure 3: Oracle score $(1 + D)/2$ for ice versus liquid, Ih versus Ic, and hexagonal versus cubic layers in stacking-disordered ice, TIP4P/Ice at 220 K. Left: thermal frames; right: the same frames after energy minimization.

Many parameters carry nearly the same information in these data. Grouping parameters whose Spearman rank correlation, pooled over thermal Ih, Ic and liquid at 220 K, exceeds 0.8 in absolute value reduces the 39 columns to 22 groups (fig. 4). The largest joins Q, the bond angle, the angular Fourier term, the three angle-histogram amplitudes, q_3, q_6 and $\hat{w}_4$: in this mixture of ambient ice and liquid they rank molecules almost identically. Pooled correlations largely reflect the ice–liquid contrast, so they show redundancy for that task, not equivalence in general, and need not hold for other phases or conditions. Other groups join $\bar{q}_3$, $\bar{q}_4$, $\bar{q}_6$ and crystalline connections; d_5, LSI, s and ζ; $\bar{Q}$ and $\bar{s}$; and the five- and six-ring fractions.

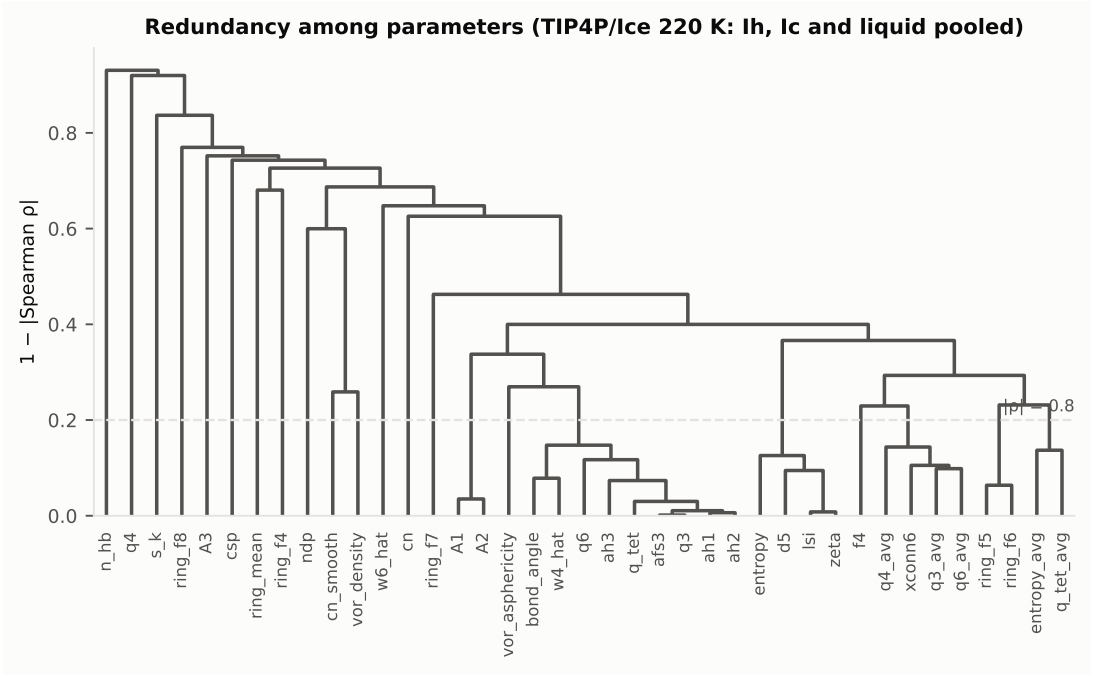


Figure 4: Hierarchical clustering of parameters by $1 - |\rho|$, with ρ the Spearman correlation over thermal Ih, Ic and liquid molecules at 220 K. The dashed line marks $|\rho| = 0.8$.

5.5 How far do thresholds transfer?

Between runs and preparations. A threshold and direction fitted on run A and applied to run B lose at most 0.008 in balanced accuracy relative to the oracle score on run B, for every parameter and population pair, thermal or quenched (fig. 5). With 10^4–10^5 molecules per population a single threshold barely overfits, and the best parameter for each task is the same under both scores. Across the five independent preparations the median standard deviation of the oracle score is 0.001; wherever the mean oracle score is at least 0.8, leaving one preparation out costs at most 0.001 and thresholds fitted on run A cost at most 0.003 (fig. 6). Hydrogen-aware parameters score within 0.009 of each other in the polarized run A crystals and the unpolarized preparations. Weak separations are another matter. In quenched stacking-disordered ice, smooth CN separates hexagonal from cubic layers at 0.60 in one preparation but 0.51–0.52 in the other four, and a threshold fitted on four preparations scores 0.45 on the fifth: the stacking sequence, not the parameter, produced the separation. In these data, every separation that failed to reproduce across preparations was below about 0.65. We use that value as an empirical warning for these datasets, not as a general reliability threshold.

What matters more than the run is which populations are compared. With the bulk liquid of run B instead of the far liquid around the seeds of run A, 25 of 156 ice/liquid scores change by more than 0.01, all of them distance- or density-based (CN, smooth CN, d_5, Voronoi, bond angle, A_1, A_2, centrosymmetry, D), by up to 0.05.

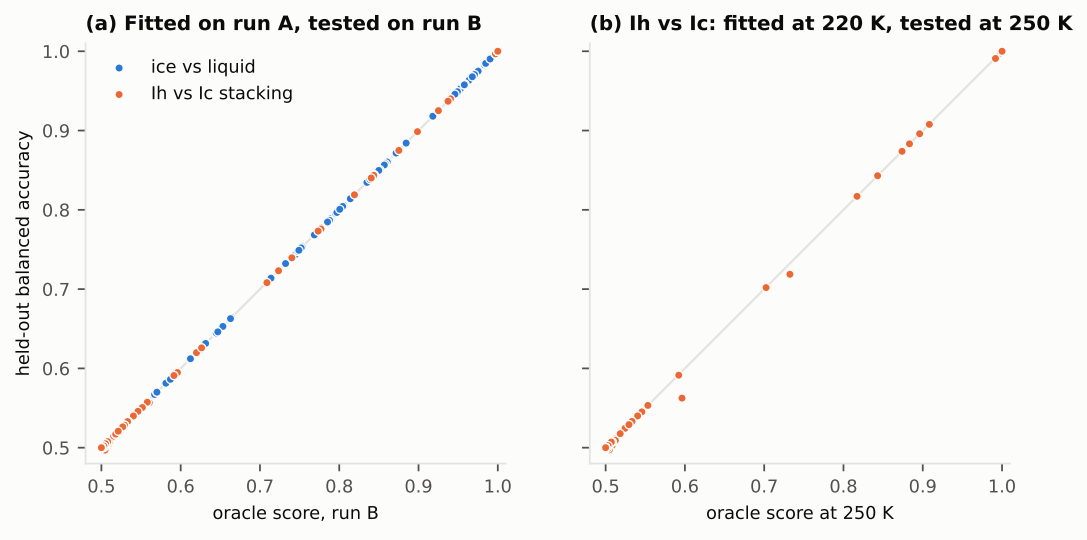


Figure 5: Held-out against oracle scores, one point per parameter and population pair (thermal and quenched). (a) Threshold fitted on run A, balanced accuracy on run B, at 220 K. (b) Ih versus Ic in run B: threshold fitted at 220 K, balanced accuracy at 250 K.

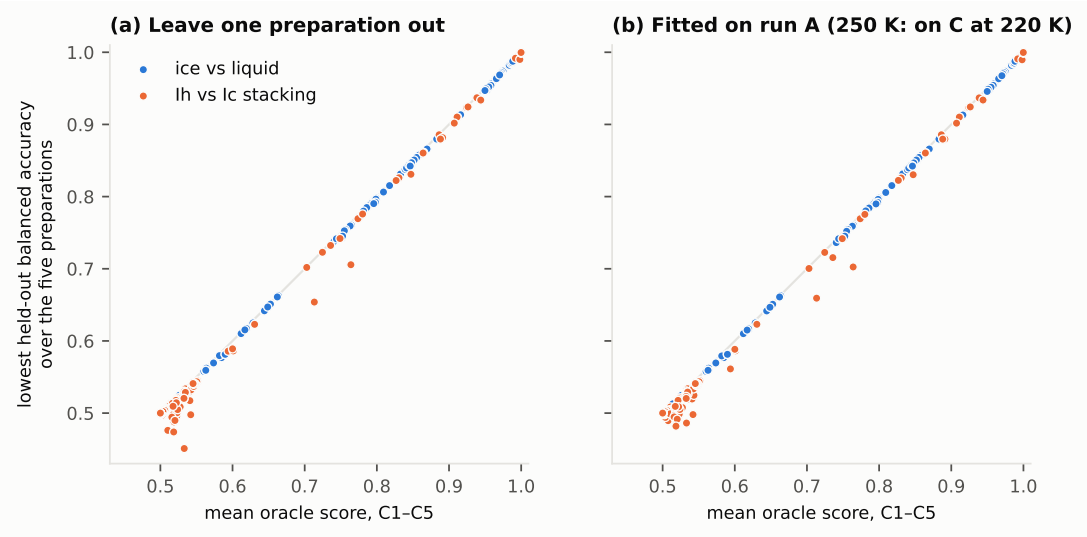


Figure 6: Five independent preparations (C1–C5). Vertical axis: lowest held-out balanced accuracy over the five preparations, for a threshold fitted (a) on the other four preparations and (b) on run A.

Between water models. Thermal frames transfer between the two TIP4P models (fig. 7a–c). From TIP4P/Ice to TIP4P/2005, oracle scores change by a median 0.004 (at most 0.025) at the same temperature or the same undercooling, and wherever the oracle score is at least 0.8 a threshold fitted on TIP4P/Ice loses at most 0.033 on the worst TIP4P/2005 preparation. Changing the protocol from the constructed constant-volume box to NPT costs at most 0.012. Quenched frames are the exception: their distributions are so narrow that the best threshold sits at a model-specific value. Smooth CN and Voronoi density keep oracle scores of 0.97 and 0.82 for ice versus liquid in either model, but thresholds moved from TIP4P/Ice score only 0.47–0.65 on TIP4P/2005.

The oxygen-only mW model is a harder test (fig. 7d). The leading parameters still transfer within 0.01: $\bar{s}$ (1.000), $\bar{q}_6$ (0.998), $\bar{Q}$ (0.991) and $\bar{q}_3$ (0.987) for ice versus liquid, $\bar{q}_3$ and $\bar{q}_4$ (at

least 0.998) for Ih versus Ic. First-shell distance parameters, however, fail to transfer even in thermal frames, because the mW liquid packs more neighbors into the first shell (median smooth CN 4.76, against 4.39 for TIP4P/Ice). Smooth CN separates ice from mW liquid with an oracle score of 0.90 but scores 0.67 with the TIP4P/Ice threshold; A_1 drops from 0.80 to 0.72 and d_5 and LSI lose about 0.04.

Density artifacts in quenched frames. The narrowness of quenched distributions also turns small density differences into spurious structure. Under NPT, Ic comes out 0.35% denser than Ih in both TIP4P models, and that alone gives Voronoi density and D oracle scores of 0.74–0.90 for quenched Ih versus Ic, against 0.52–0.55 in a fixed box. In mW, with no proton disorder to blur the lattice, quenched Ih and Ic are separated almost perfectly (at least 0.998) by smooth CN, Voronoi density and asphericity, entropy and D. Neither is a stacking signal; both are lattice-parameter signals.

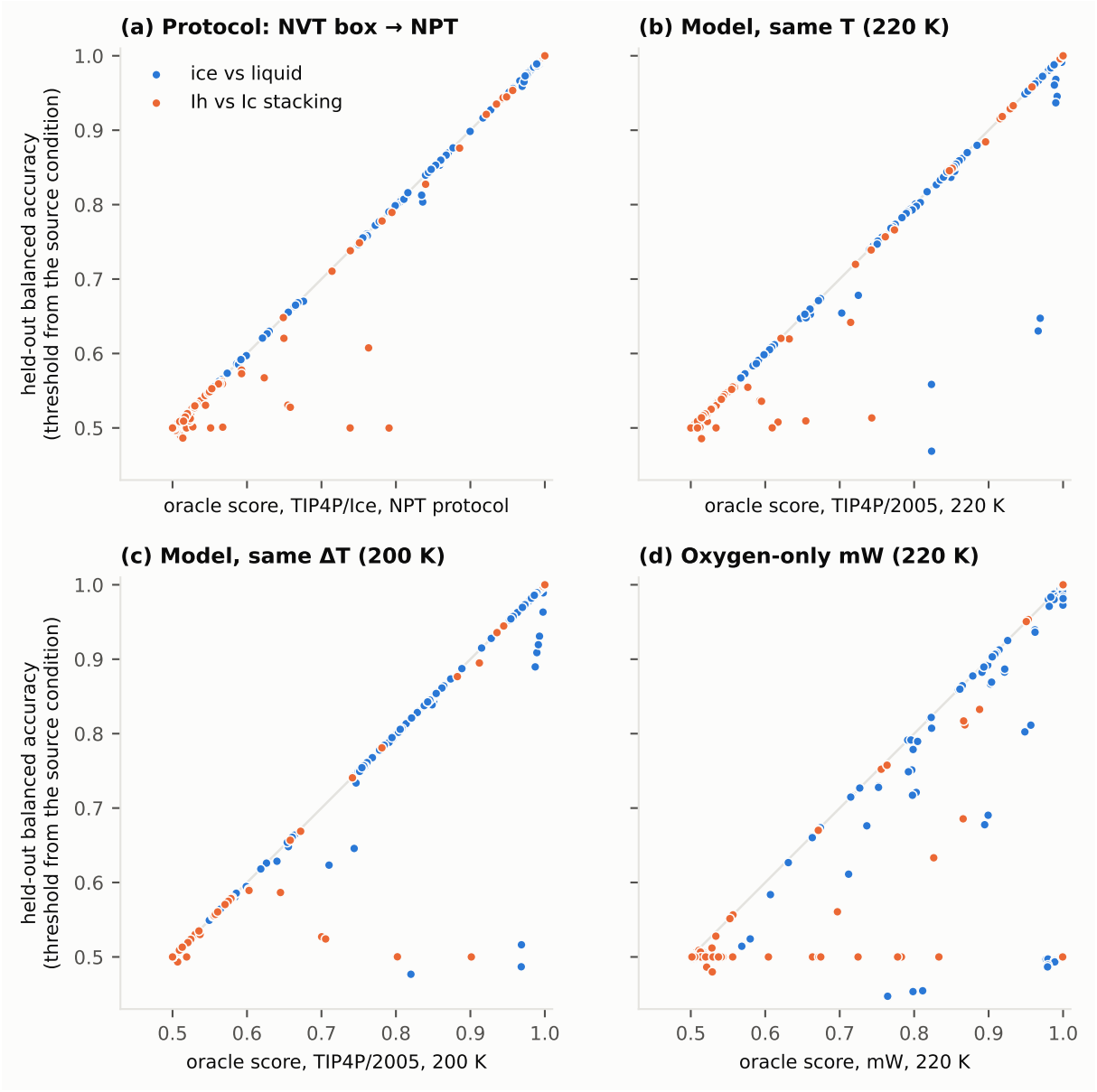


Figure 7: Moving thresholds between protocols and water models. Horizontal axis: oracle score on the target condition; vertical axis: held-out score of the threshold fitted on the source condition. (a) TIP4P/Ice, constant-volume box to NPT protocol. (b, c) TIP4P/Ice to TIP4P/2005 at 220 K and at 200 K. (d) TIP4P/Ice to mW at 220 K; hydrogen-dependent parameters are absent. The outliers far below the diagonal are distance- and density-based parameters, in quenched frames for TIP4P/2005 and also in thermal frames for mW.

5.6 High-pressure phases

Every pair of the eight thermal crystals (Ih, Ic, II, III, V, VI, VII, VIII) is still separated by some single parameter (fig. 8a): the best held-out score, taking the worse of the two preparations, is at least 0.98 for every pair except VII/VIII, where $\bar{q}_6$ reaches 0.92 (1.00 in quenched frames, through the angle histogram). Individual parameters suffer much more than this best case (fig. 8b). Of the 684 crystal pair and parameter combinations with an ideal oracle score of at least 0.95, 31% fall below 0.95 and 10% below 0.8 in thermal frames; quenching restores all but 3.5%. Losses are largest for $\hat{w}_4$ and $\hat{w}_6$ (median 0.23–0.26), D (0.14), and unaveraged q_6, F_4 and entropy

(0.06–0.08), and near zero for averaged and network parameters. $\bar{s}$ separates crystal pairs best on average (mean held-out 0.96), followed by smooth CN, Voronoi asphericity and $\bar{Q}$ (0.94).

Because the crystals were sampled at different pressures, a comparison of ice with liquid at another pressure mixes structure with state point. We therefore ran liquid at the pressure of every crystal and treat the pressure-matched pairs as the test of ice-versus-liquid identification; the other pairs measure discrimination among the sampled phase and state-point combinations. At 3.0 GPa and 220 K the liquid is nearly arrested (oxygens move 0.5 Å in 300 ps, compared with 2 Å at 0.3–0.5 GPa) and stays amorphous, with a density of 1.44 g cm^{-3} against 1.56 for ice VII, so it is closer to a compressed glass than to a liquid.

Matched against liquid at its own pressure, with the worse held-out score of the two preparations, ice VI (1.00, $\bar{s}$), VII and VIII (1.00, $\bar{s}$) are separated perfectly, II reaches 0.97 (six-ring fraction), III 0.95 ($\bar{s}$) and V only 0.84 ($\bar{q}_4$). Quenching raises V to 0.93. Ice V, with its 28-molecule cell and wide range of local environments, is the one phase that no single local parameter separates from liquid at its own pressure. No single parameter covers all matched pairs: the best worst case is $\bar{Q}$ at 0.69 (0.65 over all ice and liquid pairs). Liquids are hard to tell apart from each other at nearby pressures (0.64 between 0.3 and 0.5 GPa, 0.75 between 0.5 and 1.0 GPa, best with smooth CN). Ice-versus-liquid choices made at ambient pressure therefore do not carry over, and pairwise separation by one threshold does not imply that one classifier would sort all these phases at once. Thresholds are also fragile in quenched high-pressure frames: of the crystal pair and parameter combinations above, 62 quenched scores of at least 0.8 lose more than 0.02 between the two preparations (none in thermal frames), mostly for ices II and VIII. Quenched ordered crystals have nearly delta-shaped distributions (d_5 standard deviation 0.0002–0.003 Å), and the two preparations differ by their final box (0.4% in density for ice II), which moves the peaks by more than their width.

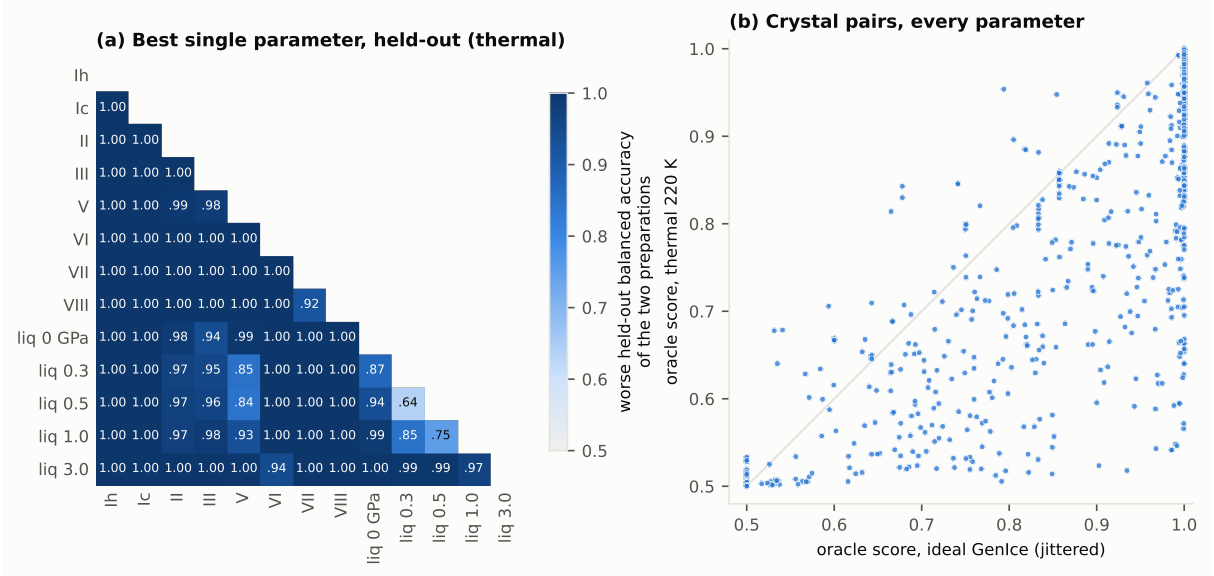


Figure 8: Thermal high-pressure phases, TIP4P/Ice at 220 K. (a) Best single parameter per pair, scored by the worse of the two held-out directions between preparations; liquid at the pressure of each crystal (0, 0.3, 0.5, 1.0 and 3.0 GPa; the 3.0 GPa liquid is nearly arrested). (b) Every parameter and crystal pair: oracle score of the ideal, jittered GenIce structures against the thermal oracle score.

5.7 Polytype recall against depth in constructed seeds

This section asks a narrower question than "does a labeler recognize ice": for molecules of a constructed Ih, Ic or stacking-disordered seed in TIP4P/Ice liquid at 220 K, how often does a labeler assign the bulk polytype of the molecule's layer, and how does that depend on the molecule's depth below the seed surface? A molecule that is not assigned the correct polytype can instead be assigned the wrong polytype, an interfacial-ice class (CHILL and CHILL+ only) or no ice class at all; these are different outcomes, and we report them separately. Each seed is one trajectory with ten frames 1 ps apart, so the intervals below describe variation within that trajectory.

In bulk ice, CHILL+ labels 99.7% of thermal Ih correctly, the original CHILL only 70.7%. The eclipsed-bond window explains the difference. In thermal bulk Ih every four-coordinated molecule keeps its three staggered bonds, but the correlation of its eclipsed bond falls inside CHILL's window $(-0.2, -0.05)$ for only 70.8% of molecules (5th–95th percentile -0.23 to 0.01), against 100% for the CHILL+ window $(-0.35, 0.25)$. After energy minimization both windows contain every eclipsed bond. This quantifies the improvement CHILL+ was designed for [108]. PTM with the default RMSD cutoff labels 97% of bulk ice correctly and never labels liquid as ice. False-positive rates in the far liquid are at most 1% for all three labelers.

In the seeds all three recover only part of the constructed crystal (fig. 9): CHILL+ finds 11% of the interior molecules of a 50-molecule Ih seed with the correct polytype, and 58% for a 200-molecule seed. Construction labels describe the starting structure, so we audited each frame before reading these numbers as errors. A seed molecule counts as intact when it lies within 1 Å of its constructed site (after removing the seed's drift) and keeps all its constructed bonds shorter than 3.5 Å. In the 100- and 200-molecule seeds 81–90% of seed molecules are intact (94–99% of interior ones); the 50-molecule Ih seed is the most deformed, 57% intact.

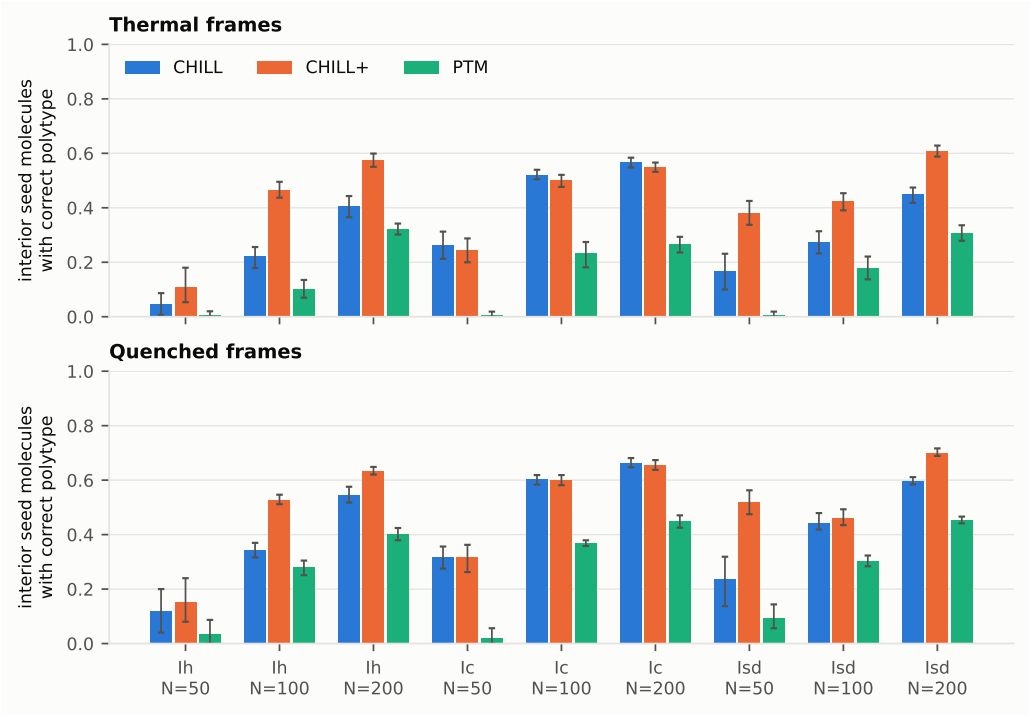


Figure 9: Fraction of interior seed molecules labeled with the correct polytype by CHILL, CHILL+ and PTM, for Ih, Ic and stacking-disordered (Isd) seeds of N molecules in TIP4P/Ice liquid at 220 K. Error bars: 95% bootstrap intervals over the ten frames of one trajectory per seed (1 ps apart); they do not include variation between independent seeds.

Polytype recall depends on depth below the constructed seed surface and hardly on seed size (fig. 10), but the three labelers behave differently. *CHILL+* assigns the correct polytype to at most 15% of molecules within 3 Å of the surface, 68–80% at 4–5 Å and at least 90% from 5–6 Å (Ic, stacking-disordered) or 6–7 Å (Ih), although about 90% of molecules are intact already at 2–4 Å. Most of the rest near the surface is not misassigned: at 3–4 Å it puts 44–47% of molecules in its interfacial-ice class, 15–25% in no ice class, and at most 7% in the wrong polytype. *CHILL* behaves like CHILL+ for Ic seeds, but for Ih and stacking-disordered seeds its recall stays at 0.5–0.9 even deep inside, the same eclipsed-window deficit as in bulk Ih. *PTM* has no interfacial class and never assigns the wrong polytype; it needs about 2 Å more depth than CHILL+, reaching 50–55% at 5–6 Å and 82–90% at 6–8 Å. A 50-molecule seed (radius about 7.3 Å) has only 3–4% of its molecules deeper than 5 Å, which explains most of its low score. Small-seed recall therefore measures how much crystalline support a labeler needs before it assigns a bulk polytype, not whether it detects ice; with its interfacial class, CHILL+ detects some form of ice at much smaller depths. The depths quoted hold for these constructed seeds in TIP4P/Ice at 220 K. This is consistent with the observation that labelers disagree mainly at interfaces [15], and it bears directly on nucleus sizes from seeding, which depend on where the labeler places the surface [19, 20, 21].

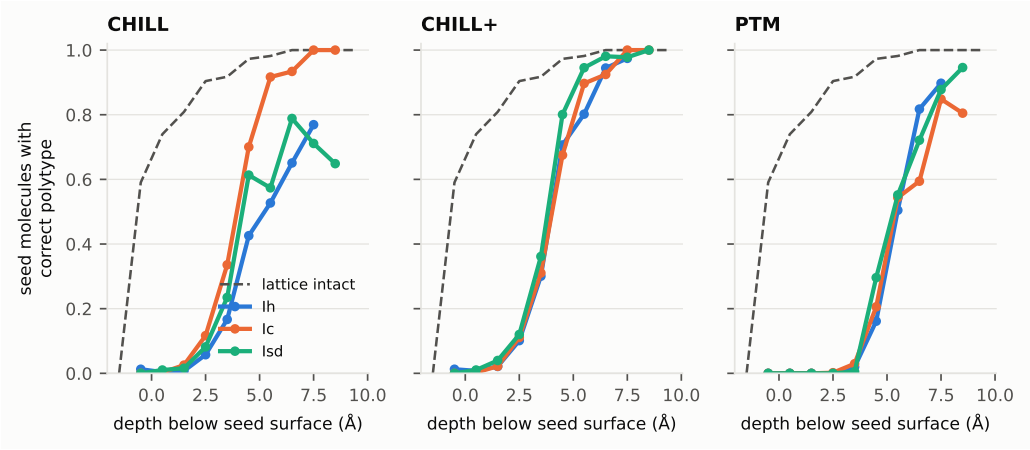


Figure 10: Polytype recall against depth: fraction of seed molecules labeled with the correct polytype against depth below the constructed seed surface, pooled over seed sizes, thermal frames at 220 K; molecules deeper than about 9 Å come only from the 200-molecule seeds. Dashed: fraction of molecules whose constructed lattice is intact. Bins with fewer than 30 molecules are omitted.

5.8 Application: nuclei along nucleation paths

As a qualitative check outside the benchmark data, we computed all parameters on 366 frames from 44 nucleation trajectories (biased path sampling, metadynamics, shooting and brute-force runs, TIP4P/Ice and mW). In the 34 frames of four fixed-box TIP4P/Ice path-sampling trajectories whose largest connected CHILL+ ice core holds at least 20 molecules, we compared each parameter with membership of that core. The median ROC area under the curve is highest for $\bar{q}_6$ (0.985), followed by $\bar{Q}$ and $\bar{q}_3$ (0.97), $\bar{s}$ (0.96, lower inside the core), crystalline connections (0.95) and $\bar{q}_4$ (0.93); N_{HB} is near 0.5. The ranking matches the bulk benchmark. Because CHILL+ is itself built from $\mathbf{q}_3$ correlations, this is agreement with a reference labeler, not independent accuracy, and the biased paths give no rates.

5.9 Sensitivity to the neighbor convention

All results above use the registry defaults. To see how much they depend on that choice, we varied one factor at a time on preparations C1–C3 at 220 K (bulk Ih, bulk Ic and liquid, 12 thermal frames 25 ps apart and their quenched counterparts; 571,392 environments): the first-shell cutoff of every cutoff-based parameter and of CHILL and CHILL+ from 3.2 to 3.8 Å, the LSI shell radius over the same range, the averaging radius of $\bar{q}_l$ and $\bar{Q}$ from 3.2 to 5.0 Å with q_{lm} and Q kept at 3.5 Å, and the integration radius r_m of the pair entropy, which in our implementation is also its averaging radius, from 4.0 to 6.0 Å. Rank-based parameters (d_5, Q, S_k, centrosymmetry) and Voronoi parameters have no cutoff. Besides the oracle score we fitted a threshold on C1 at the default setting and applied it to C2 and C3 computed with the varied setting, which is what happens when a published threshold is reused with a different convention (fig. 11).

First-shell cutoff. The leading parameters barely notice it. For thermal Ih versus liquid, $\bar{q}_6$ stays at 0.987–0.992 and the six-ring fraction at 0.989–0.992; for Ih versus Ic, $\bar{q}_3$ stays at 0.983–1.000, $\bar{q}_4$ at 0.993–0.997, F_4 at 0.930–0.939 and crystalline connections at 0.906–0.925, all lowest at 3.8 Å. Their thresholds fitted at 3.5 Å lose at most 0.025 at any other cutoff. The

ranking of the 30 scanned columns is stable (Spearman correlation with the default ranking at least 0.94 for ice versus liquid and 0.99 for Ih versus Ic), and the three best parameters for Ih versus Ic are the same at every cutoff. Unaveraged first-shell parameters are another story. For thermal Ih versus liquid, 15 of the 30 columns change by more than 0.02, all of them rising with the cutoff: CN from 0.53 to 0.81, smooth CN from 0.63 to 0.87, D from 0.57 to 0.76, the bond-angle parameter from 0.73 to 0.85 and A_1 from 0.76 to 0.87. A larger cutoff reaches the interstitial molecules that distinguish the liquid, so none of these parameters has a natural cutoff, and published values computed with different cutoffs cannot be compared. Smooth CN is the extreme case: its threshold fitted with $r_0 = 3.5$ Å scores only 0.50–0.69 at any other r_0. Hydrogen-bond parameters (N_{HB}, ζ, ring fractions) and the LSI shell radius hardly matter (LSI 0.832–0.834), because the angular part of the hydrogen-bond criterion, not the distance, decides which pairs count.

Averaging radius. Averaging further than the first shell improves the even-l parameters: for thermal Ih versus liquid, $\bar{Q}$ rises from 0.976 to 1.000 and $\bar{q}_4$ from 0.870 to 0.923 at 5 Å, and for Ih versus Ic $\bar{q}_6$ rises from 0.776 to 0.962. $\bar{q}_3$ does the opposite. Its perfect Ih/Ic separation exists only with first-shell averaging and falls to 0.894, 0.647 and 0.804 at averaging radii of 4.0, 4.5 and 5.0 Å. The $\mathbf{q}_3$ vectors of neighboring molecules in a tetrahedral network are anti-correlated (the staggered bonds of CHILL), so the average depends on how many molecules of each sublattice it contains, and the second shell adds them in a different proportion. Thresholds of $\bar{q}_3$, and of $\bar{q}_4$ for ice versus liquid, do not survive a change of averaging radius at all (0.50).

Entropy radius. $\bar{s}$ separates thermal ice from liquid at 0.95 with $r_m = 4$ Å and 1.00 from 5 Å on; it never separates thermal Ih from Ic (at most 0.63). In quenched frames it separates Ih from Ic perfectly at the default $r_m = 5.0$ Å but only at 0.53–0.77 at any other radius: the integration limit happens to fall where the sharp quenched radial distributions of the two polytypes differ. The perfect quenched separation in fig. 3 is an accident of the cutoff, not a property of the parameter. Thresholds of s and $\bar{s}$ are tied to r_m (0.50 when it changes).

Labelers. CHILL+ labels 99.7% of thermal bulk Ih correctly with cutoffs of 3.3–3.5 Å, 98.6% at 3.7 Å and 96.1% at 3.8 Å, because it labels only molecules with exactly four neighbors inside the cutoff. CHILL uses the four nearest neighbors and is insensitive to the cutoff (70.7–71.0% for Ih, at least 99.3% for Ic). Both keep false positives in the liquid below 1% at every cutoff.

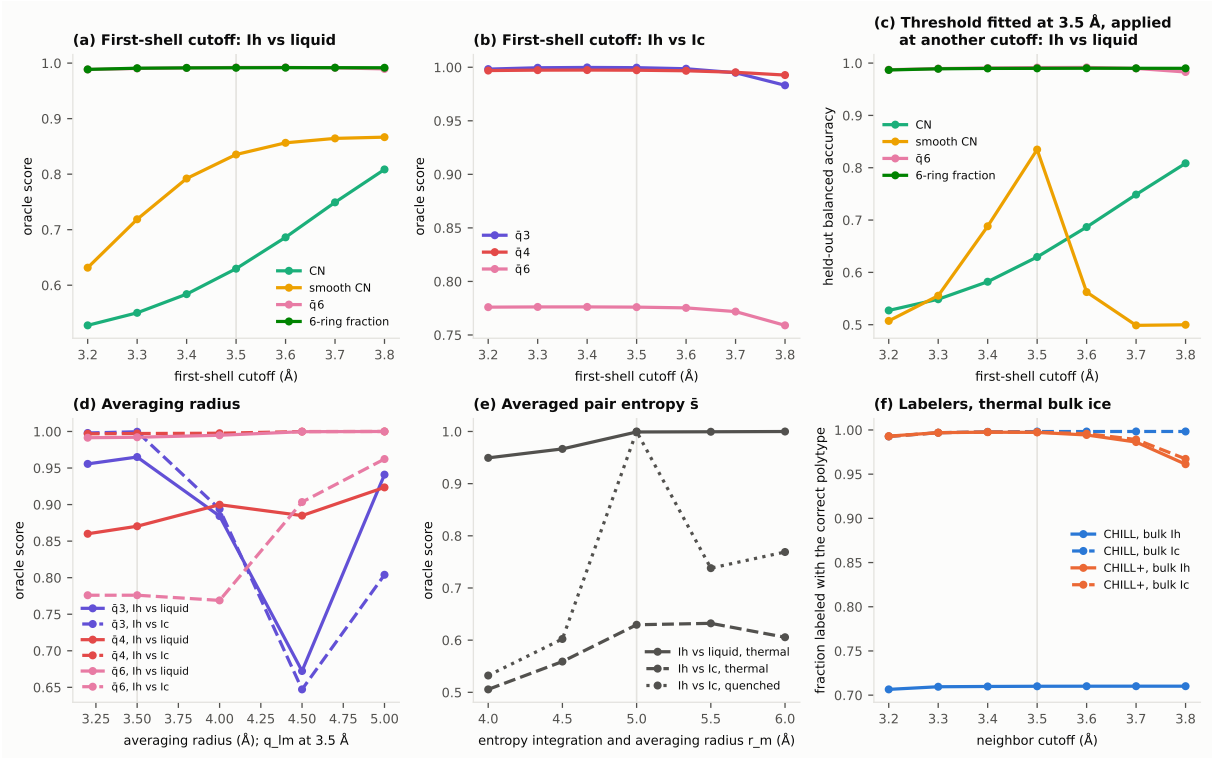


Figure 11: Sensitivity to the neighbor convention, run C1–C3 at 220 K. (a, b) Oracle score against the first-shell cutoff of every cutoff-based parameter ($\bar{q}_6$ and the six-ring fraction overlap near 0.99 in (a)). (c) Held-out score of a threshold fitted on C1 at 3.5 Å and applied to C2 and C3 computed with another cutoff (worse of the two). (d) Averaging radius of $\bar{q}_l$, with q_{lm} at 3.5 Å. (e) Integration and averaging radius of $\bar{s}$. (f) Fraction of thermal bulk ice labeled with the correct polytype by CHILL and CHILL+. Thin vertical lines mark the defaults.

5.10 Computational cost

We timed every registry entry on periodic tilings of one thermal frame of C1 (Ih: 1920 to 61,440 molecules; liquid: 4096 to 65,536) on one Intel Core i7-12700F, starting each measurement from a fresh analysis snapshot so that the neighbor search is included, as in the benchmark itself (fig. 12). Every entry scales linearly with system size (fitted exponents 0.99–1.04; PTM is sublinear because of a fixed setup cost). On one thread the whole registry costs about 140 μs per molecule, 9.2 s for a 65,536-molecule frame. The cheapest entries are CN (1.8 μs per molecule), PTM (2.7 μs) and CHILL and CHILL+ (5 μs); most others cost 6–13 μs, largely the neighbor search to 5 Å. The most expensive are ring statistics (30 μs), the angle histogram (21 μs), Voronoi (20 μs), A_1–A_3 (19 μs) and the pair entropy (18 μs). Cost is therefore not a reason to prefer a weaker parameter: the recommended $\bar{s}$, $\bar{q}_l$ and $\bar{Q}$ cost 8–18 μs per molecule, a negligible fraction of the simulation that produced the frame.

Threads help less than one might expect. With eight threads, ring statistics run 3.2 times faster, crystalline connections 2.5 times and Steinhardt parameters 2.4 times, but the full registry is no faster (0.94 times the serial speed at 65,536 molecules, and slower for small systems), because neighbor construction and several kernels run serially, and the common-neighborhood parameters, whose loop holds the Python interpreter lock, become four times slower. For trajectories, parallelism over frames is the effective choice; the benchmark itself ran ten frames at a time in separate processes.

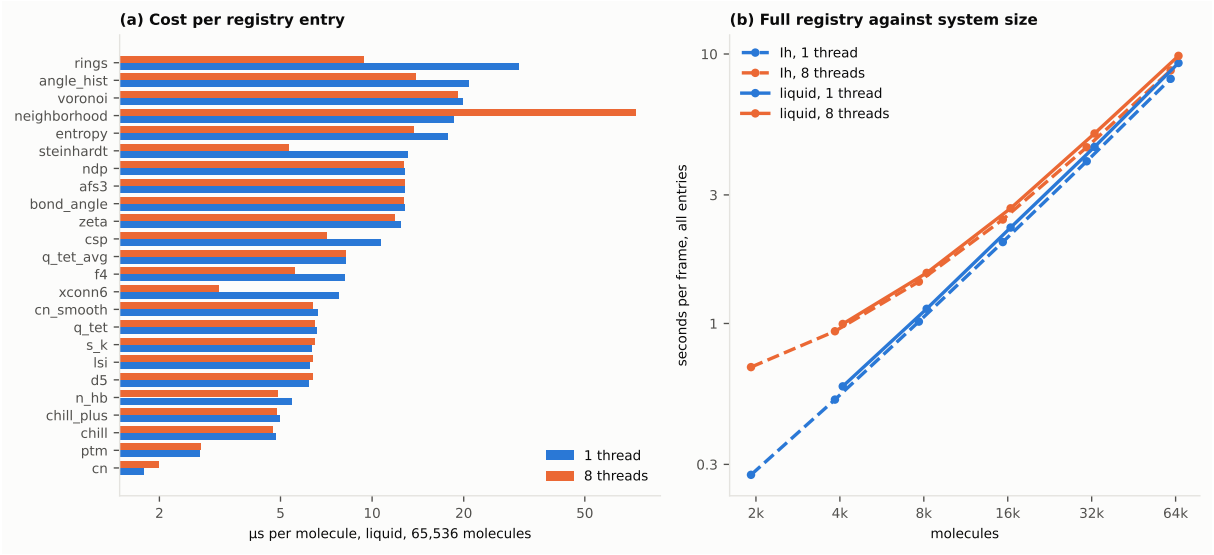


Figure 12: Computational cost on an Intel Core i7-12700F, median of three warm repeats, each from a fresh snapshot (neighbor search included). (a) Time per molecule for each registry entry, liquid with 65,536 molecules, one and eight threads. (b) Time per frame for all entries against system size; Ih solid, liquid dashed.

5.11 What the benchmark does not cover

Table 3 defines the boundaries of the benchmark. Within them, several questions remain open. The sensitivity scan covers one temperature and the ambient phases only, and varies one factor at a time. Only one temperature is available for most phases, and one pressure for each high-pressure phase. Ices IV, IX and XI–XVII and the empty clathrate frameworks were studied only as ideal structures; amorphous ices were not studied at all, and no three-site water model was included. Pairwise separation by one threshold does not establish that a single multiclass classifier would work for all phases at once. The supercooled liquid references are still aging slowly, and ideal-crystal results for proton-ordered phases depend on the GenIce cell geometries. Finally, scoring per molecule does not test a parameter as a CV; that requires continuity at neighbor exchanges, correct gradients and, ideally, committor analysis.

6 Practical guide

Table 4 condenses the benchmark into recommendations by task. Five rules of thumb follow from it.

1. *Average.* At finite temperature, averaging over the first shell is worth more than any other design choice. Averaged entropy, $\bar{q}_6$ and $\bar{Q}$ separate ambient ice from liquid at 0.98–1.00; their unaveraged forms at 0.78–0.85.

2. *Use bond orientations for stacking.* Only averaged low-l Steinhardt parameters, F_4 and crystalline connections see the difference between Ih and Ic in thermal frames; rings and radial parameters do not.

3. *Do not move thresholds of distance- or density-based parameters* between water models,

boxes, protocols or cutoffs without recalibrating them, and be suspicious of separations found only in quenched frames.

4. *Keep the averaging radius with the threshold.* The first-shell cutoff barely affects averaged and bond-orientational parameters, but the averaging radius and the entropy radius change their values and, for $\bar{q}_3$, their meaning. Report both and recalibrate when either changes.

5. *Recalibrate for high pressure.* No single parameter separates all high-pressure ices from liquid at their own pressure; choose by pair, and compare against liquid at the same pressure.

Table 4: Recommendations, valid for the conditions of table 3: TIP4P/Ice at 220 K (Ih/Ic also at 250 K) with five independent preparations; TIP4P/2005 at 220 and 200 K and mW at 220 K with three; six high-pressure ices and liquids at 0.3–3.0 GPa and 220 K with two; default neighbor settings unless stated. Rankings are the same for oracle and held-out scores. In these data, separations below about 0.65 did not reproduce across preparations.

Question	Best in the benchmark	Avoid
Ice vs. liquid, ambient	$\bar{s}$, $\bar{q}_6$, $\bar{Q}$; six-ring fraction if no averaging is wanted	CN, N_{HB}
Ih vs. Ic	$\bar{q}_3$, $\bar{q}_4$; F_4 if hydrogens are available	ring statistics, unaveraged parameters
Polytype labels in nuclei	CHILL+ (Ih, stacking-disordered), CHILL or CHILL+ (Ic); in constructed seeds at 220 K, polytype recall reaches 90% only 5–7 Å below the surface, and CHILL+ assigns shallower molecules mostly to its interfacial class	CHILL for Ih; PTM polytype labels within ∼7 Å of an interface
Distinguishing crystal phases	$\bar{s}$, smooth CN, Voronoi asphericity, $\bar{Q}$; $\bar{q}_6$ for VII/VIII	$\hat{w}_4$, $\hat{w}_6$, D, N_{HB}, A_3
High-pressure ice vs. liquid at the same pressure	per pair (fig. 8): $\bar{s}$ for III, VI, VII, VIII; six-ring fraction for II; $\bar{q}_4$ for V (0.84 at best); no single parameter above 0.69 for all	ambient-pressure choices without checking
Proton-ordered vs. disordered	no local parameter tested is reliable (VII/VIII separate through their oxygen lattices)	–
Oxygen-only models (mW)	$\bar{s}$, $\bar{q}_6$, $\bar{Q}$, $\bar{q}_3$ (ice vs. liquid) and $\bar{q}_3$, $\bar{q}_4$ (Ih vs. Ic); their TIP4P/Ice thresholds transferred within 0.01 at 220 K, thermal frames, default settings	thresholds of distance parameters from other models

6.1 Software

Most parameters are available in more than one package, but with different conventions. freud [156] and pyscal [157] compute Steinhardt parameters (including averaged forms) and Voronoi tessellations; OVITO [158] provides PTM, identify-diamond, centrosymmetry and CHILL+; PLUMED [159, 160] provides differentiable versions of coordination and Steinhardt-type parameters, and with mlcolvar [142] learned CVs; MDAnalysis [161, 162] handles trajectories and hydrogen-bond analysis; DScribe [125] computes high-dimensional descriptors. SolOrder, used here, implements all parameters of table 2 on MDAnalysis universes with parallel, Numba-compiled kernels, and wraps SOAP, ACSF and MBTR through DScribe. Its thread parallelism

pays off only for a few parameters (section 5.10). It is an analysis library: none of its parameters provides gradients, so CVs should be taken from PLUMED. Packages differ in neighbor rules, normalization and the averaging of q_{lm} vectors versus scalars; agreement between them should be checked, not assumed.

6.2 Reproducibility checklist

When reporting results that depend on a local order parameter, state:

1. the exact definition and the code and version used;
2. the neighbor rule (cutoff, number of neighbors or Voronoi), the averaging neighborhood, and whether hydrogens are used;
3. the water model and state point, and where that state point lies on the model's own phase diagram;
4. the threshold, how it was calibrated (on which reference populations, thermal or quenched), and whether it was tested on held-out data;
5. for labelers, the classes reported (including interfacial classes) and how clusters are built from labels;
6. for CVs, how continuity and gradients were checked.

7 Outlook

Four problems stand out. First, proton order. No local parameter we tested separates proton-ordered from proton-disordered forms of one oxygen framework, and those that seem to do so use the accompanying lattice distortion. Polarization and longer-range orientational correlations remain to be tested as local observables. Second, interfaces. Every labeler we tested needs several Ångströms of crystal before it assigns a molecule its bulk polytype, and labelers differ in what they call the remaining near-surface molecules, which makes the size of small nuclei and the width of interfaces depend on the labeler. A benchmark with ground truth at interfaces, for example from constructed seeds of many sizes, would test learned classifiers where they are needed. Third, transferability. Thresholds of averaged and bond-orientational parameters move between water models; thresholds of distance-based parameters do not, and learned models trained on one model should be tested on others, including machine-learned potentials of first-principles quality. Fourth, collective variables. Good per-molecule separation does not make a good CV; parameters need smooth neighbor membership and verified gradients, and their quality as reaction coordinates should be judged by committor analysis rather than by separation of the end states. Amorphous ices, dense liquids at high pressure and confined water are the natural next datasets.

Data and code availability

SolOrder (version 1.1.0, MIT license), the benchmark scripts, settings registry and result tables are available in the SolOrder repository. The figures in section 5 are generated by

`benchmarks/figures.py` from the CSV files in `benchmarks/results`.

References

[1] Christoph G. Salzmann. Advances in the experimental exploration of water's phase diagram. *The Journal of Chemical Physics*, 150(6), feb 2019.

[2] Thomas C. Hansen. The everlasting hunt for new ice phases. *Nature Communications*, 12(1), May 2021.

[3] Thomas Loerting, Violeta Fuentes-Landete, Christina M. Tonauer, and Tobias M. Gasser. Open questions on the structures of crystalline water ices. *Communications Chemistry*, 3(1):109, 2020.

[4] Tobias M. Gasser, Alexander V. Thoeny, A. Dominic Fortes, and Thomas Loerting. Structural characterization of ice xix as the second polymorph related to ice vi. *Nature Communications*, 12(1):1128, 2021.

[5] Ryo Yamane, Kazuki Komatsu, Jun Gouchi, Yoshiya Uwatoko, Shinichi Machida, Takanori Hattori, Hayate Ito, and Hiroyuki Kagi. Experimental evidence for the existence of a second partially-ordered phase of ice vi. *Nature Communications*, 12(1):1129, 2021.

[6] Yun-Hee Lee, Jin Kyun Kim, Yong-Jae Kim, Minju Kim, Yong Chan Cho, Rachel J. Husband, Cornelius Strohm, Emma Ehrenreich-Petersen, Konstantin Glazyrin, Torsten Laurus, Heinz Graafsma, Robert P. C. Bauer, Felix Lehmkühler, Karen Appel, Zuzana Konôpková, Minxue Tang, Anand Prashant Dwivedi, Jolanta Sztuck-Dambietz, Lisa Randolph, Khachiwan Buakor, Oliver Humphries, Carsten Baehtz, Tobias Eklund, Lisa Katharina Mohrbach, Anshuman Mondal, Hauke Marquardt, Earl Francis O'Bannon, Katrin Amann-Winkel, Choong-Shik Yoo, Ulf Zastrau, Hanns-Peter Liermann, Hiroki Nada, and Geun Woo Lee. Multiple freezing–melting pathways of high-density ice through ice xxi phase at room temperature. *Nature Materials*, 25(2):302–309, 2025.

[7] Israel Osmond, Mikhail A. Kuzovnikov, David A. Lewis, Hannah A. Shuttleworth, Tomás Marqueño, Milo J. Dixon, Cerian E. A. Robertson, Vitan Cherepnalkoski, Bihan Wang, Andreas Hermann, Miriam Peña-Alvarez, and Ross T. Howie. Ultrahigh pressure phase transition in ice. *Nature Materials*, 25(10):1850–1854, 2026.

[8] Gabriele C. Sosso, Ji Chen, Stephen J. Cox, Martin Fitzner, Philipp Pedevilla, Andrea Zen, and Angelos Michaelides. Crystal nucleation in liquids: Open questions and future challenges in molecular dynamics simulations. *Chemical Reviews*, 116(12):7078–7116, 2016.

[9] E. Sanz, C. Vega, J. R. Espinosa, R. Caballero-Bernal, J. L. F. Abascal, and C. Valeriani. Homogeneous ice nucleation at moderate supercooling from molecular simulation. *Journal of the American Chemical Society*, 135(40):15008–15017, 2013.

[10] Amir Haji-Akbari and Pablo G. Debenedetti. Direct calculation of ice homogeneous nucleation rate for a molecular model of water. *Proceedings of the National Academy of Sciences*, 112(34):10582–10588, 2015.

[11] Laura Lupi, Arpa Hudait, Baron Peters, Michael Grünwald, Ryan Gotchy Mullen, Andrew H. Nguyen, and Valeria Molinero. Role of stacking disorder in ice nucleation. *Nature*, 551(7679):218–222, 2017.

[12] Pablo M. Piaggi, Jack Weis, Athanassios Z. Panagiotopoulos, Pablo G. Debenedetti, and Roberto Car. Homogeneous ice nucleation in an ab initio machine-learning model of water. *Proceedings of the National Academy of Sciences*, 119(33):e2207294119, 2022.

[13] Emily B. Moore and Valeria Molinero. Is it cubic? ice crystallization from deeply supercooled water. *Physical Chemistry Chemical Physics*, 13(44):20008, 2011.

[14] Tamsin L. Malkin, Benjamin J. Murray, Andrey V. Brukhno, Jamshed Anwar, and Christoph G. Salzmann. Structure of ice crystallized from supercooled water. *Proceedings of the National Academy of Sciences*, 109(4):1041–1045, jan 2012.

[15] Jihong Shi, Maxwell Fulford, Hui Li, Mariam Marzook, Maryam Reisjalali, Matteo Salvalaglio, and Carla Molteni. Investigating the quasi-liquid layer on ice surfaces: a comparison of order parameters. *Physical Chemistry Chemical Physics*, 24(20):12476–12487, 2022.

[16] A. Moradzadeh, H. Oliaei, and N. R. Aluru. Topology-Based Phase Identification of Bulk, Interface, and Confined Water Using an Edge-Conditioned Convolutional Graph Neural Network. *The Journal of Physical Chemistry C*, 127(5):2612–2621, 2023.

[17] Tarak Karmakar, Michele Invernizzi, Valerio Rizzi, and Michele Parrinello. Collective variables for the study of crystallisation. *Molecular Physics*, 119(19-20):e1893848, 2021.

[18] Neha, Vikas Tiwari, Soumya Mondal, Nisha Kumari, and Tarak Karmakar. Collective Variables for Crystallization Simulations–from Early Developments to Recent Advances. *ACS Omega*, 8(1):127–146, 2022.

[19] Jorge R. Espinosa, Carlos Vega, Chantal Valeriani, and Eduardo Sanz. Seeding approach to crystal nucleation. *The Journal of Chemical Physics*, 144(3):034501, 2016.

[20] Ignacio Sanchez-Burgos, Andres R. Tejedor, Carlos Vega, Maria M. Conde, Eduardo Sanz, Jorge Ramirez, and Jorge R. Espinosa. Homogeneous ice nucleation rates for mw and tip4p/ice models through lattice mold calculations. *The Journal of Chemical Physics*, 157(9):094503, 2022.

[21] Willem Gispen, Jorge R. Espinosa, Eduardo Sanz, Carlos Vega, and Marjolein Dijkstra. Variational umbrella seeding for calculating nucleation barriers. *The Journal of Chemical Physics*, 160(17):174501, 2024.

[22] Katarina E. Blow, David Quigley, and Gabriele C. Sosso. The seven deadly sins: When computing crystal nucleation rates, the devil is in the details. *The Journal of Chemical Physics*, 155(4):040901, 2021.

[23] Elise Duboué-Dijon and Damien Laage. Characterization of the local structure in liquid water by various order parameters. *The Journal of Physical Chemistry B*, 119(26):8406–8418, 2015.

[24] Edward Danquah Donkor, Alessandro Laio, and Ali Hassanali. Do machine-learning atomic descriptors and order parameters tell the same story? the case of liquid water. *Journal of Chemical Theory and Computation*, 19(14):4596–4605, March 2023.

[25] Ryan S. DeFever, Colin Targonski, Steven W. Hall, Melissa C. Smith, and Sapna Sarupria. A generalized deep learning approach for local structure identification in molecular simulations. *Chemical Science*, 10(32):7503–7515, 2019.

[26] QHwan Kim, Joon-Hyuk Ko, Sunghoon Kim, and Wonho Jhe. GCIceNet: a graph convolutional network for accurate classification of water phases. *Physical Chemistry Chemical Physics*, 22(45):26340–26350, 2020.

[27] Hideo Doi, Kazuaki Z. Takahashi, and Takeshi Aoyagi. Searching local order parameters to classify water structures of ice ih, ic, and liquid. *The Journal of Chemical Physics*, 154(16), April 2021.

[28] Hideo Doi, Kazuaki Z. Takahashi, and Takeshi Aoyagi. Mining of effective local order parameters to classify ice polymorphs. *The Journal of Physical Chemistry A*, 125(43):9518–9526, 2021.

[29] Hideo Doi, Kazuaki Z. Takahashi, and Takeshi Aoyagi. Searching for local order parameters to classify water structures at triple points. *Journal of Computational Chemistry*, 42(24):1720–1727, 2021.

[30] Tobias M. Gasser, Alexander V. Thoeny, Christina Tonauer, Johannes Bachler, Violeta Fuentes-Landete, and Thomas Loerting. *How Many Crystalline Ices are There?*, pages 105–129. CRC Press, 2022.

[31] Christoph G. Salzmann, John S. Loveday, Alexander Rosu-Finsen, and Craig L. Bull. Structure and nature of ice xix. *Nature Communications*, 12(1):3162, 2021.

[32] Hiroki Kobayashi, Kazuki Komatsu, Kenji Mochizuki, Hayate Ito, Koichi Momma, Shinichi Machida, Takanori Hattori, Kunio Hirata, Yoshiaki Kawano, Saori Maki-Yonekura, Kiyofumi Takaba, Koji Yonekura, Qianli Xue, Misaki Sato, and Hiroyuki Kagi. New metastable ice phases via supercooled water, 2025. Preprint.

[33] J. L. Aragones, M. M. Conde, E. G. Noya, and C. Vega. The phase diagram of water at high pressures as obtained by computer simulations of the tip4p/2005 model: the appearance of a plastic crystal phase. *Phys. Chem. Chem. Phys.*, 11(3):543–555, 2009.

[34] Maria Rescigno, Alberto Toffano, Umbertoluca Ranieri, Leon Andriambariarijaona, Richard Gaal, Stefan Klotz, Michael Marek Koza, Jacques Ollivier, Fausto Martelli, John Russo, Francesco Sciortino, Jose Teixeira, and Livia Eleonora Bove. Observation of plastic ice vii by quasi-elastic neutron scattering. *Nature*, 640(8059):662–667, 2025.

[35] Marius Millot, Sebastien Hamel, J. Ryan Rygg, Peter M. Celliers, Gilbert W. Collins, Federica Coppari, Dayne E. Fratanduono, Raymond Jeanloz, Damian C. Swift, and Jon H. Eggert. Experimental evidence for superionic water ice using shock compression. *Nature Physics*, 14(3):297–302, feb 2018.

[36] Marius Millot, Federica Coppari, J. Ryan Rygg, Antonio Correa Barrios, Sebastien Hamel, Damian C. Swift, and Jon H. Eggert. Nanosecond x-ray diffraction of shock-compressed superionic water ice. *Nature*, 569(7755):251–255, 2019.

[37] Vitali B. Prakapenka, Nicholas Holtgrewe, Sergey S. Lobanov, and Alexander F. Goncharov. Structure and properties of two superionic ice phases. *Nature Physics*, 17(11):1233–1238, oct 2021.

[38] Bingqing Cheng, Mandy Bethkenhagen, Chris J. Pickard, and Sebastien Hamel. Phase behaviours of superionic water at planetary conditions. *Nature Physics*, 17(11):1228–1232, 2021.

[39] Christoph G. Salzmann, Paolo G. Radaelli, Andreas Hallbrucker, Erwin Mayer, and John L. Finney. The preparation and structures of hydrogen ordered phases of ice. *Science*, 311(5768):1758–1761, mar 2006.

[40] Christoph G. Salzmann, Paolo G. Radaelli, Erwin Mayer, and John L. Finney. Ice XV: A new thermodynamically stable phase of ice. *Physical Review Letters*, 103(10), sep 2009.

[41] Christoph G. Salzmann, Alexander Rosu-Finsen, Zainab Sharif, Paolo G. Radaelli, and John L. Finney. Detailed crystallographic analysis of the ice v to ice XIII hydrogen-ordering phase transition. *The Journal of Chemical Physics*, 154(13), apr 2021.

[42] Tamsin L. Malkin, Benjamin J. Murray, Christoph G. Salzmann, Valeria Molinero, Steven J. Pickering, and Thomas F. Whale. Stacking disorder in ice i. *Physical Chemistry Chemical Physics*, 17(1):60–76, 2015.

[43] Werner F. Kuhs, Christian Sippel, Andrzej Falenty, and Thomas C. Hansen. Extent and relevance of stacking disorder in "ice i_c". *Proceedings of the National Academy of Sciences*, 109(52):21259–21264, 2012.

[44] Leonardo del Rosso, Milva Celli, Francesco Grazzi, Michele Catti, Thomas C. Hansen, A. Dominic Fortes, and Lorenzo Ulivi. Cubic ice ic without stacking defects obtained from ice XVII. *Nature Materials*, 19(6):663–668, feb 2020.

[45] Kazuki Komatsu, Shinichi Machida, Fumiya Noritake, Takanori Hattori, Asami Sano-Furukawa, Ryo Yamane, Keishiro Yamashita, and Hiroyuki Kagi. Ice ic without stacking disorder by evacuating hydrogen from hydrogen hydrate. *Nature Communications*, 11(1):464, 2020.

[46] Haiyang Niu, Yi Isaac Yang, and Michele Parrinello. Temperature Dependence of Homogeneous Nucleation in Ice. *Physical Review Letters*, 122(24):245501, 2019.

[47] E. Sanz, C. Vega, J. L. F. Abascal, and L. G. MacDowell. Phase diagram of water from computer simulation. *Physical Review Letters*, 92(25):255701, 2004.

[48] C. Vega, E. Sanz, and J. L. F. Abascal. The melting temperature of the most common models of water. *The Journal of Chemical Physics*, 122(11):114507, 2005.

[49] C. Vega, J. L. F. Abascal, M. M. Conde, and J. L. Aragones. What ice can teach us about water interactions: a critical comparison of the performance of different water models. *Faraday Discuss.*, 141:251–276, 2009.

[50] Carlos Vega and Jose L. F. Abascal. Simulating water with rigid non-polarizable models: a general perspective. *Physical Chemistry Chemical Physics*, 13(44):19663, 2011.

[51] J. L. F. Abascal, E. Sanz, R. García Fernández, and C. Vega. A potential model for the study of ices and amorphous water: Tip4p/ice. *The Journal of Chemical Physics*, 122(23):234511, 2005.

[52] Sigbjørn L. Bore, Pablo M. Piaggi, Roberto Car, and Francesco Paesani. Phase diagram of the TIP4p/ice water model by enhanced sampling simulations. *The Journal of Chemical Physics*, 157(5), August 2022.

[53] J. L. F. Abascal and C. Vega. A general purpose model for the condensed phases of water: Tip4p/2005. *The Journal of Chemical Physics*, 123(23):234505, 2005.

[54] M. M. Conde, M. Rovere, and P. Gallo. High precision determination of the melting points of water tip4p/2005 and water tip4p/ice models by the direct coexistence technique. *The Journal of Chemical Physics*, 147(24):244506, 2017.

[55] Masanori Hirata, Takuma Yagasaki, Masakazu Matsumoto, and Hideki Tanaka. Phase diagram of TIP4p/2005 water at high pressure. *Langmuir*, 33(42):11561–11569, August 2017.

[56] Valeria Molinero and Emily B. Moore. Water modeled as an intermediate element between carbon and silicon. *The Journal of Physical Chemistry B*, 113(13):4008–4016, 2009.

[57] Bingqing Cheng, Edgar A. Engel, Jörg Behler, Christoph Dellago, and Michele Ceriotti. Ab initio thermodynamics of liquid and solid water. *Proceedings of the National Academy of Sciences*, 116(4):1110–1115, January 2019.

[58] Aleks Reinhardt and Bingqing Cheng. Quantum-mechanical exploration of the phase diagram of water. *Nature Communications*, 12(1):588, 2021.

[59] Linfeng Zhang, Han Wang, Roberto Car, and Weinan E. Phase diagram of a deep potential water model. *Physical Review Letters*, 126(23):236001, 2021.

[60] Sigbjørn Løland Bore and Francesco Paesani. Realistic phase diagram of water from "first principles" data-driven quantum simulations. *Nature Communications*, 14(1):3349, 2023.

[61] Eslam Ibrahim, Yury Lysogorskiy, Ralf Drautz, and Pablo Piaggi. Water phase diagram from a general-purpose atomic cluster expansion potential, 2026. Preprint.

[62] Masakazu Matsumoto, Takuma Yagasaki, and Hideki Tanaka. Genice: Hydrogen-disordered ice generator. *Journal of Computational Chemistry*, 39(1):61–64, 2018.

[63] Masakazu Matsumoto, Takuma Yagasaki, and Hideki Tanaka. Genice-core: Efficient algorithm for generation of hydrogen-disordered ice structures. *The Journal of Chemical Physics*, 160(9):094101, 2024.

[64] Thomas Loerting, Katrin Winkel, Markus Seidl, Marion Bauer, Christian Mitterdorfer, Philip H. Handle, Christoph G. Salzmann, Erwin Mayer, John L. Finney, and Daniel T. Bowron. How many amorphous ices are there? *Physical Chemistry Chemical Physics*, 13(19):8783, 2011.

[65] Katrin Amann-Winkel, Roland Böhmer, Franz Fujara, Catalin Gainaru, Burkhard Geil, and Thomas Loerting. Colloquium: Water's controversial glass transitions. *Reviews of Modern Physics*, 88(1):011002, 2016.

[66] Philip H. Handle, Thomas Loerting, and Francesco Sciortino. Supercooled and glassy water: Metastable liquid(s), amorphous solid(s), and a no-man's land. *Proceedings of the National Academy of Sciences*, 114(51):13336–13344, 2017.

[67] Christina M. Tonauer, Lilli-Ruth Fidler, Johannes Giebelmann, Keishiro Yamashita, and Thomas Loerting. Nucleation and growth of crystalline ices from amorphous ices. *The Journal of Chemical Physics*, 158(14):141001, 2023.

[68] Alexander Rosu-Finsen, Michael B. Davies, Alfred Amon, Han Wu, Andrea Sella, Angelos Michaelides, and Christoph G. Salzmann. Medium-density amorphous ice. *Science*, 379(6631):474–478, 2023.

[69] Andrzej Falenty, Thomas C. Hansen, and Werner F. Kuhs. Formation and properties of ice XVI obtained by emptying a type sII clathrate hydrate. *Nature*, 516(7530):231–233, dec 2014.

[70] Leonardo del Rosso, Milva Celli, and Lorenzo Ulivi. New porous water ice metastable at atmospheric pressure obtained by emptying a hydrogen-filled ice. *Nature Communications*, 7(1), nov 2016.

[71] John Russo and Hajime Tanaka. Understanding water's anomalies with locally favoured structures. *Nature Communications*, 5(1):3556, 2014.

[72] K. T. Wikfeldt, A. Nilsson, and L. G. M. Pettersson. Spatially inhomogeneous bimodal inherent structure of simulated liquid water. *Physical Chemistry Chemical Physics*, 13(44):19918, 2011.

[73] Hajime Tanaka, Hua Tong, Rui Shi, and John Russo. Revealing key structural features hidden in liquids and glasses. *Nature Reviews Physics*, 1(5):333–348, May 2019.

[74] Liwen Li, Jie Zhong, Jun Zhang, Zhiyuan Wang, and Xiao Cheng Zeng. Evidence for the generic existence of two local structures in liquid water. *Nature Physics*, 22(7):1144–1151, 2026.

[75] Bartomeu Monserrat, Jan Gerit Brandenburg, Edgar A. Engel, and Bingqing Cheng. Liquid water contains the building blocks of diverse ice phases. *Nature Communications*, 11(1), November 2020.

[76] Riccardo Capelli, Francesco Muniz-Miranda, and Giovanni M. Pavan. Ephemeral ice-like local environments in classical rigid models of liquid water. *The Journal of Chemical Physics*, 156(21), June 2022.

[77] Walter Mickel, Sebastian C. Kapfer, Gerd E. Schröder-Turk, and Klaus Mecke. Shortcomings of the bond orientational order parameters for the analysis of disordered particulate matter. *The Journal of Chemical Physics*, 138(4), January 2013.

[78] Wolfgang Lechner and Christoph Dellago. Accurate determination of crystal structures based on averaged local bond order parameters. *The Journal of Chemical Physics*, 129(11), September 2008.

[79] Ivan Saika-Voivod, Francesco Sciortino, and Peter H. Poole. Computer simulations of liquid silica: equation of state and liquid–liquid phase transition. *Physical Review E*, 63(1), December 2000.

[80] Megan J. Cuthbertson and Peter H. Poole. Mixturelike behavior near a liquid-liquid phase transition in simulations of supercooled water. *Physical Review Letters*, 106(11), March 2011.

[81] Eli Shiratani and Masaki Sasai. Growth and collapse of structural patterns in the hydrogen bond network in liquid water. *The Journal of Chemical Physics*, 104(19):7671–7680, May 1996.

[82] Eli Shiratani and Masaki Sasai. Molecular scale precursor of the liquid–liquid phase transition of water. *The Journal of Chemical Physics*, 108(8):3264–3276, February 1998.

[83] N. N. Medvedev, V. P. Voloshin, and Yu. I. Naberukhin. Structure of simple liquids as a problem of percolation in the voronoi grid. *Journal of Structural Chemistry*, 30(2):253–260, 1989.

[84] G. Ruocco, M. Sampoli, and R. Vallauri. Analysis of the network topology in liquid water and hydrogen sulphide by computer simulation. *The Journal of Chemical Physics*, 96(8):6167–6176, April 1992.

[85] Pál Jedlovszky. Voronoi polyhedra analysis of the local structure of water from ambient to supercritical conditions. *The Journal of Chemical Physics*, 111(13):5975–5985, oct 1999.

[86] Pablo M. Piaggi and Michele Parrinello. Entropy based fingerprint for local crystalline order. *The Journal of Chemical Physics*, 147(11), September 2017.

[87] Pablo M. Piaggi, Omar Valsson, and Michele Parrinello. Enhancing entropy and enthalpy fluctuations to drive crystallization in atomistic simulations. *Physical Review Letters*, 119(1):015701, 2017.

[88] P.-L. CHAU and A. J. HARDWICK. A new order parameter for tetrahedral configurations. *Molecular Physics*, 93(3):511–518, feb 1998.

[89] Jeffrey R. Errington and Pablo G. Debenedetti. Relationship between structural order and the anomalies of liquid water. *Nature*, 409(6818):318–321, January 2001.

[90] Hideo Doi, Kazuaki Z. Takahashi, and Takeshi Aoyagi. Mining of effective local order parameters for classifying crystal structures: A machine learning study. *The Journal of Chemical Physics*, 152(21):214501, 2020.

[91] Kazuaki Z. Takahashi. Molecular cluster analysis using local order parameters selected by machine learning. *Physical Chemistry Chemical Physics*, 25(1):658–672, 2023.

[92] G. J. Ackland and A. P. Jones. Applications of local crystal structure measures in experiment and simulation. *Physical Review B*, 73(5), February 2006.

[93] Albert P. Bartók, Risi Kondor, and Gábor Csányi. On representing chemical environments. *Physical Review B*, 87(18), May 2013.

[94] J. Dana. Honeycutt and Hans C. Andersen. Molecular dynamics study of melting and freezing of small lennard-jones clusters. *The Journal of Physical Chemistry*, 91(19):4950–4963, September 1987.

[95] Ali Radhi and Kamran Behdinan. Identification of crystal structures in atomistic simulation by predominant common neighborhood analysis. *Computational Materials Science*, 126:182–190, January 2017.

[96] Alenka Luzar and David Chandler. Effect of environment on hydrogen bond dynamics in liquid water. *Physical Review Letters*, 76(6):928–931, February 1996.

[97] P.M. Rodger, T.R. Forester, and W. Smith. Simulations of the methane hydrate/methane gas interface near hydrate forming conditions conditions. *Fluid Phase Equilibria*, 116(1-2):326–332, March 1996.

[98] Felipe Jiménez-Ángeles and Abbas Firoozabadi. Nucleation of methane hydrates at moderate subcooling by molecular dynamics simulations. *The Journal of Physical Chemistry C*, 118(21):11310–11318, May 2014.

[99] SHIRLEY V. KING. Ring configurations in a random network model of vitreous silica. *Nature*, 213(5081):1112–1113, 1967.

[100] D. S. Franzblau. Computation of ring statistics for network models of solids. *Physical Review B*, 44(10):4925–4930, 1991.

[101] M. Matsumoto, A. Baba, and I. Ohmine. Topological building blocks of hydrogen bond network in water. *The Journal of Chemical Physics*, 127(13):134504, 2007.

[102] Maud Formanek and Fausto Martelli. Probing the network topology in network-forming materials: The case of water. *AIP Advances*, 10(5):055205, 2020.

[103] Paul J. Steinhardt, David R. Nelson, and Marco Ronchetti. Bond-orientational order in liquids and glasses. *Physical Review B*, 28(2):784–805, 1983.

[104] Jorge Benet, Luis G. MacDowell, and Eduardo Sanz. A study of the ice–water interface using the tip4p/2005 water model. *Phys. Chem. Chem. Phys.*, 16(40):22159–22166, 2014.

[105] Pieter-Rein ten Wolde, Maria J. Ruiz-Montero, and Daan Frenkel. Simulation of homogeneous crystal nucleation close to coexistence. *Faraday Discussions*, 104:93, 1996.

[106] Aleks Reinhardt, Jonathan P. K. Doye, Eva G. Noya, and Carlos Vega. Local order parameters for use in driving homogeneous ice nucleation with all-atom models of water. *The Journal of Chemical Physics*, 137(19), November 2012.

[107] Emily B. Moore, Ezequiel de la Llave, Kai Welke, Damian A. Scherlis, and Valeria Molinero. Freezing, melting and structure of ice in a hydrophilic nanopore. *Physical Chemistry Chemical Physics*, 12(16):4124, 2010.

[108] Andrew H. Nguyen and Valeria Molinero. Identification of clathrate hydrates, hexagonal ice, cubic ice, and liquid water in simulations: the CHILL+ algorithm. *The Journal of Physical Chemistry B*, 119(29):9369–9376, November 2014.

[109] Cynthia L. Kelchner, S. J. Plimpton, and J. C. Hamilton. Dislocation nucleation and defect structure during surface indentation. *Physical Review B*, 58(17):11085–11088, November 1998.

[110] Peter Mahler Larsen, Søren Schmidt, and Jakob Schiøtz. Robust structural identification via polyhedral template matching. *Modelling and Simulation in Materials Science and Engineering*, 24(5):055007, May 2016.

[111] E. Maras, O. Trushin, A. Stukowski, T. Ala-Nissila, and H. Jónsson. Global transition path search for dislocation formation in ge on si(001). *Computer Physics Communications*, 205:13–21, August 2016.

[112] Alexander Stukowski. Structure identification methods for atomistic simulations of crystalline materials. *Modelling and Simulation in Materials Science and Engineering*, 20(4):045021, May 2012.

[113] Fausto Martelli, Hsin-Yu Ko, Erdal C. Oğuz, and Roberto Car. Local-order metric for condensed-phase environments. *Physical Review B*, 97(6):064105, 2018.

[114] Fausto Martelli, Nicolas Giovambattista, Salvatore Torquato, and Roberto Car. Searching for crystal-ice domains in amorphous ices. *Physical Review Materials*, 2(7):075601, 2018.

[115] Masakazu Matsumoto, Takuma Yagasaki, and Hideki Tanaka. A bayesian approach for identification of ice Ih, ice Ic, high density, and low density liquid water with a torsional order parameter. *The Journal of Chemical Physics*, 150(21):214504, 2019.

[116] Riccardo Foffi and Francesco Sciortino. Identification of local structures in water from supercooled to ambient conditions. *The Journal of Chemical Physics*, 160(9):094504, 2024.

[117] Pablo M. Piaggi and Roberto Car. Phase equilibrium of liquid water and hexagonal ice from enhanced sampling molecular dynamics simulations. *The Journal of Chemical Physics*, 152(20):204116, 2020.

[118] Kazuaki Z. Takahashi and Masaki Hiratsuka. Local order parameters classifying water networks of ice and cyclopropane clathrate hydrates. *Crystal Growth & Design*, 23(7):4815–4824, 2023.

[119] Satoki Ishiai, Katsuhiro Endo, and Kenji Yasuoka. Graph neural networks classify molecular geometry and design novel order parameters of crystal and liquid. *The Journal of Chemical Physics*, 159(6):064103, 2023.

[120] Felix Musil, Andrea Grisafi, Albert P. Bartók, Christoph Ortner, Gábor Csányi, and Michele Ceriotti. Physics-Inspired Structural Representations for Molecules and Materials. *Chemical Reviews*, 121(16):9759–9815, 2021.

[121] Marcel F. Langer, Alex Goeßmann, and Matthias Rupp. Representations of molecules and materials for interpolation of quantum-mechanical simulations via machine learning. *npj Computational Materials*, 8(1):41, 2022.

[122] Jörg Behler. Atom-centered symmetry functions for constructing high-dimensional neural network potentials. *The Journal of Chemical Physics*, 134(7), February 2011.

[123] Michael J. Willatt, Félix Musil, and Michele Ceriotti. Atom-density representations for machine learning. *The Journal of Chemical Physics*, 150(15):154110, 2019.

[124] Haoyan Huo and Matthias Rupp. Unified representation of molecules and crystals for machine learning. *Machine Learning: Science and Technology*, 3(4):045017, November 2022.

[125] Lauri Himanen, Marc O.J. Jäger, Eiaki V. Morooka, Filippo Federici Canova, Yashasvi S. Ranawat, David Z. Gao, Patrick Rinke, and Adam S. Foster. DScribe: Library of descriptors for machine learning in materials science. *Computer Physics Communications*, 247:106949, February 2020.

[126] Adu Offei-Danso, Ali Hassanali, and Alex Rodriguez. High-dimensional fluctuations in liquid water: Combining chemical intuition with unsupervised learning. *Journal of Chemical Theory and Computation*, 18(5):3136–3150, April 2022.

[127] Philipp Geiger and Christoph Dellago. Neural networks for local structure detection in polymorphic systems. *The Journal of Chemical Physics*, 139(16):164105, 2013.

[128] Maxwell Fulford, Matteo Salvalaglio, and Carla Molteni. DeepIce: A deep neural network approach to identify ice and water molecules. *Journal of Chemical Information and Modeling*, 59(5):2141–2149, March 2019.

[129] Hong Sun, Sebastien Hamel, Tim Hsu, Babak Sadigh, Vincenzo Lordi, and Fei Zhou. Ice phase classification made easy with score-based denoising. *Journal of Chemical Information and Modeling*, 64(16):6369–6376, 2024.

[130] Dibyendu Maity and Suman Chakrabarty. IceCoder: Identification of ice phases in molecular simulation using variational autoencoder. *Journal of Chemical Theory and Computation*, 21(4):1916–1928, 2025.

[131] Emanuele Boattini, Marjolein Dijkstra, and Laura Filion. Unsupervised learning for local structure detection in colloidal systems. *The Journal of Chemical Physics*, 151(15):154901, 2019.

[132] Emanuele Boattini, Susana Marín-Aguilar, Saheli Mitra, Giuseppe Foffi, Frank Smallenburg, and Laura Filion. Autonomously revealing hidden local structures in supercooled liquids. *Nature Communications*, 11(1):5479, 2020.

[133] Sébastien Becker, Emilie Devijver, Rémi Molinier, and Noël Jakse. Unsupervised topological learning for identification of atomic structures. *Physical Review E*, 105(4):045304, 2022.

[134] Sébastien Becker, Emilie Devijver, Rémi Molinier, and Noël Jakse. Unsupervised topological learning approach of crystal nucleation. *Scientific Reports*, 12(1):3195, 2022.

[135] Edgar A. Engel, Andrea Anelli, Michele Ceriotti, Chris J. Pickard, and Richard J. Needs. Mapping uncharted territory in ice from zeolite networks to ice structures. *Nature Communications*, 9(1):2173, 2018.

[136] Edward Danquah Donkor, Adu Offei-Danso, Alex Rodriguez, Francesco Sciortino, and Ali Hassanali. Beyond Local Structures in Critical Supercooled Water through Unsupervised Learning. *The Journal of Physical Chemistry Letters*, 15(15):3996–4005, 2024.

[137] D. Quigley and P. M. Rodger. Metadynamics simulations of ice nucleation and growth. *The Journal of Chemical Physics*, 128(15):154518, 2008.

[138] Yue-Yu Zhang, Haiyang Niu, GiovanniMaria Piccini, Dan Mendels, and Michele Parrinello. Improving collective variables: The case of crystallization. *The Journal of Chemical Physics*, 150(9):094509, 2019.

[139] Luigi Bonati, Valerio Rizzi, and Michele Parrinello. Data-Driven Collective Variables for Enhanced Sampling. *The Journal of Physical Chemistry Letters*, 11(8):2998–3004, 2020.

[140] Luigi Bonati, GiovanniMaria Piccini, and Michele Parrinello. Deep learning the slow modes for rare events sampling. *Proceedings of the National Academy of Sciences*, 118(44):e2113533118, 2021.

[141] Michele Invernizzi and Michele Parrinello. Rethinking Metadynamics: From Bias Potentials to Probability Distributions. *The Journal of Physical Chemistry Letters*, 11(7):2731–2736, 2020.

[142] Luigi Bonati, Enrico Trizio, Andrea Rizzi, and Michele Parrinello. A unified framework for machine learning collective variables for enhanced sampling simulations: mlcolvar. *The Journal of Chemical Physics*, 159(1):014801, 2023.

[143] Florian M. Dietrich, Xavier R. Advincula, Gianpaolo Gobbo, Michael A. Bellucci, and Matteo Salvalaglio. Machine Learning Nucleation Collective Variables with Graph Neural Networks. *Journal of Chemical Theory and Computation*, 20(4):1600–1611, 2023.

[144] Jintu Zhang, Luigi Bonati, Enrico Trizio, Odin Zhang, Yu Kang, TingJun Hou, and Michele Parrinello. Descriptor-Free Collective Variables from Geometric Graph Neural Networks. *Journal of Chemical Theory and Computation*, 20(24):10787–10797, 2024.

[145] Linfeng Zhang, Jiequn Han, Han Wang, Roberto Car, and Weinan E. Deep Potential Molecular Dynamics: A Scalable Model with the Accuracy of Quantum Mechanics. *Physical Review Letters*, 120(14):143001, 2018.

[146] Volodymyr Babin, Claude Leforestier, and Francesco Paesani. Development of a "First Principles" Water Potential with Flexible Monomers: Dimer Potential Energy Surface, VRT Spectrum, and Second Virial Coefficient. *Journal of Chemical Theory and Computation*, 9(12):5395–5403, 2013.

[147] Volodymyr Babin, Gregory R. Medders, and Francesco Paesani. Development of a "First Principles" Water Potential with Flexible Monomers. II: Trimer Potential Energy Surface, Third Virial Coefficient, and Small Clusters. *Journal of Chemical Theory and Computation*, 10(4):1599–1607, 2014.

[148] Gregory R. Medders, Volodymyr Babin, and Francesco Paesani. Development of a "First-Principles" Water Potential with Flexible Monomers. III. Liquid Phase Properties. *Journal of Chemical Theory and Computation*, 10(8):2906–2910, 2014.

[149] Christoph Schran, Fabian L. Thiemann, Patrick Rowe, Erich A. Müller, Ondrej Marsalek, and Angelos Michaelides. Machine learning potentials for complex aqueous systems made simple. *Proceedings of the National Academy of Sciences*, 118(38):e2110077118, 2021.

[150] Bingqing Cheng, Christoph Dellago, and Michele Ceriotti. Theoretical prediction of the homogeneous ice nucleation rate: disentangling thermodynamics and kinetics. *Physical Chemistry Chemical Physics*, 20(45):28732–28740, 2018.

[151] Haiyang Niu, Luigi Bonati, Pablo M. Piaggi, and Michele Parrinello. Ab initio phase diagram and nucleation of gallium. *Nature Communications*, 11(1):2654, 2020.

[152] Pablo M. Piaggi, Annabella Selloni, Athanassios Z. Panagiotopoulos, Roberto Car, and Pablo G. Debenedetti. A first-principles machine-learning force field for heterogeneous ice nucleation on microcline feldspar. *Faraday Discussions*, 249:98–113, 2024.

[153] Steven W. Rick and A. D. J. Haymet. Dielectric constant and proton order and disorder in ice ih: Monte carlo computer simulations. *The Journal of Chemical Physics*, 118(20):9291–9296, 2003.

[154] V. Buch, P. Sandler, and J. Sadlej. Simulations of h_2o solid, liquid, and clusters, with an emphasis on ferroelectric ordering transition in hexagonal ice. *The Journal of Physical Chemistry B*, 102(44):8641–8653, 1998.

[155] Peter Eastman, Raimondas Galvelis, Raúl P. Peláez, Charlles R. A. Abreu, Stephen E. Farr, Emilio Gallicchio, Anton Gorenko, Michael M. Henry, Frank Hu, Jing Huang, Andreas Krämer, Julien Michel, Joshua A. Mitchell, Vijay S. Pande, João PGLM Rodrigues, Jaime Rodriguez-Guerra, Andrew C. Simmonett, Sukrit Singh, Jason Swails, Philip Turner, Yuanqing Wang, Ivy Zhang, John D. Chodera, Gianni De Fabritiis, and Thomas E. Markland. OpenMM 8: Molecular Dynamics Simulation with Machine Learning Potentials. *The Journal of Physical Chemistry B*, 128(1):109–116, 2023.

[156] Vyas Ramasubramani, Bradley D. Dice, Eric S. Harper, Matthew P. Spellings, Joshua A. Anderson, and Sharon C. Glotzer. freud: A software suite for high throughput analysis of particle simulation data. *Computer Physics Communications*, 254:107275, 2020.

[157] Sarath Menon, Grisell Leines, and Jutta Rogal. pyscal: A python module for structural analysis of atomic environments. *Journal of Open Source Software*, 4(43):1824, 2019.

[158] Alexander Stukowski. Visualization and analysis of atomistic simulation data with OVITO–the Open Visualization Tool. *Modelling and Simulation in Materials Science and Engineering*, 18(1):015012, 2009.

[159] Gareth A. Tribello, Massimiliano Bonomi, Davide Branduardi, Carlo Camilloni, and Giovanni Bussi. PLUMED 2: New feathers for an old bird. *Computer Physics Communications*, 185(2):604–613, 2014.

[160] The PLUMED consortium. Promoting transparency and reproducibility in enhanced molecular simulations. *Nature Methods*, 16(8):670–673, 2019.

[161] Naveen Michaud-Agrawal, Elizabeth J. Denning, Thomas B. Woolf, and Oliver Beckstein. MDAnalysis: A toolkit for the analysis of molecular dynamics simulations. *Journal of Computational Chemistry*, 32(10):2319–2327, 2011.

[162] Richard Gowers, Max Linke, Jonathan Barnoud, Tyler Reddy, Manuel Melo, Sean Seyler, Jan Domański, David Dotson, Sébastien Buchoux, Ian Kenney, and Oliver Beckstein. MDAnalysis: A Python Package for the Rapid Analysis of Molecular Dynamics Simulations. In *Proceedings of the 15th Python in Science Conference*, pages 98–105. SciPy, 2016.